

Supplementary Information Note 3

MRMhub Data Processing Pipeline for Dataset 1

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Prerequisites

Run this code only to install or update `midar`. If you already have the latest version skip this step.

```
#if (!require("remotes")) install.packages("remotes")
#remotes::install_github("SLINGhub/MRMhub", subdir = "quant")
```

Data Processing and Quality Control Pipeline

1. Load the `mrmhub` package

```
library(mrmhub)
```

2. Import the analysis data (V1)

After creating a `MidarExperiment` object to store all our data, we importing the `MRMkit` results containing peak areas (`intensity`), retention time (`rt`), and peak full width at half maximum (`fwhm`). Available metadata, i.e., acquisition time stamp and precursor and product `m/z` values are imported as well (`import_metadata = TRUE`).

```

data_path <- "./data/dataset-1/data/sPerfect_LongCirc_MRMkit_Initial.csv"
mexp_initial <- MRMhubExperiment()
mexp_initial <- import_data_mrmhub(mexp_initial, data_path, import_metadata = TRUE)
#> v Imported 937 analyses with 503 features
#> i `feature_area` selected as default feature intensity. Modify with `set_intensity_var()`
#> v Analysis metadata associated with 937 analyses.
#> v Feature metadata associated with 503 features.

```

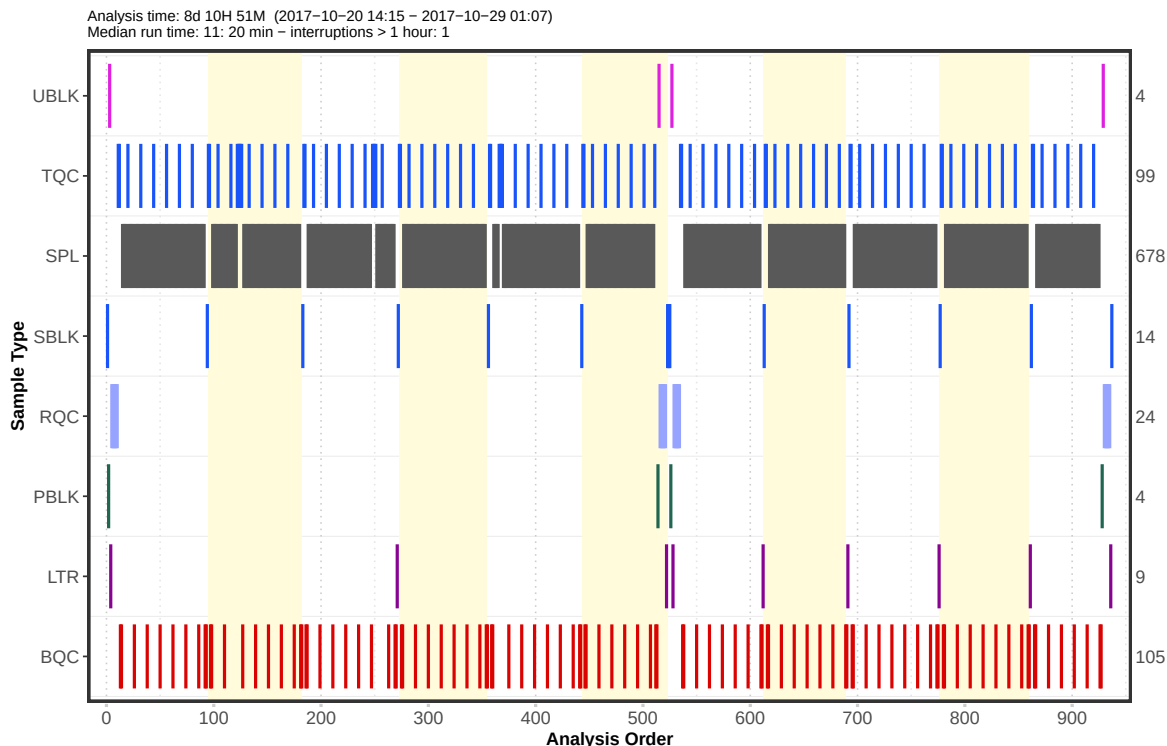
3. Analytical design and timeline

The plot below provides an overview of the sample and batch structure, the quality control (QC) samples included with their respective positions with, and information regarding the date, duration, and run time of the analysis.

```

fig4a <- plot_runsequence(
  mexp_initial,
  show_batches = TRUE,
  qc_types = c("SPL", "BQC", "TQC", "PBLK", "UBLK", "RQC", "SBLK", "LTR"),
  batch_zebra_stripe = TRUE, base_font_size = 6,
  batch_fill_color = "#fffbdb",
  segment_linewidth = 0.5,
  show_timestamp = FALSE) +
theme(plot.title = element_text(size = 5))
fig4a

```



4. Overall trends and possible outliers

To examine overall technical trends and issues affecting the majority of analytes (features), the RLA (Relative Log Abundance) plot is a useful tool (De Livera et al., Analytical Chemistry, 2015). In this plot, all features are normalised (by across or within-batch medians) and plotted as a boxplot per sample. This plot can help to identify potential technical issues, such as pipetting errors, sample loss, changes in injection volume or instrument sensitivity.

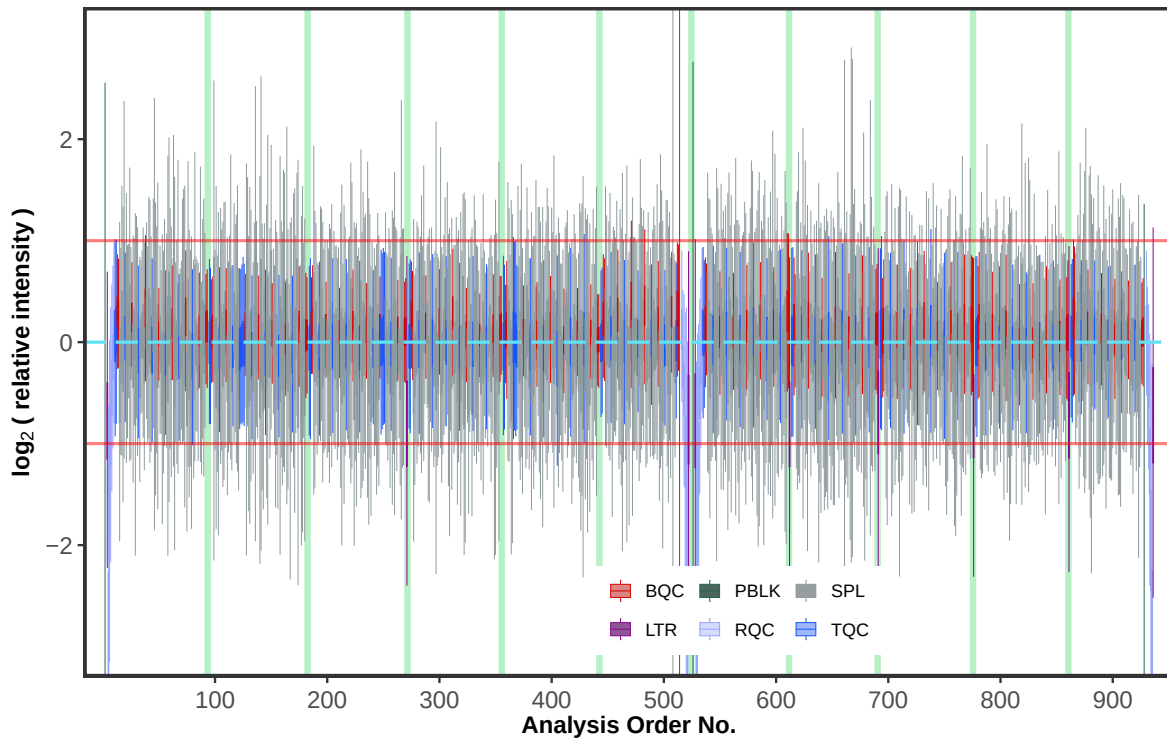
First an overview of all analyses (runs) are plotted

```
plot_rla_boxplot(
  data = mexp_initial,
  rla_type_batch = c("within"),
  variable = "intensity",
  qc_types = c("BQC", "SPL", "LTR", "TQC", "RQC", "PBLK"),
  filter_data = FALSE,
  #analysis_range = get_batch_boundaries(myexp, batch_indices = c(5,6)),
  y_lim = c(-3,3),
  show_timestamp = FALSE,
  outlier_method = "fold",
```

```

outlier_k = c(-1, 1),
outlier_exclude = FALSE,
x_gridlines = FALSE,
batch_zebra_stripe = FALSE,
linewidth = 0.1, show_plot = TRUE)
#> i Found 1 outliers in the 919 shown analyses

```



A zoom in batch-by-batch reveals that one QC/sample type, the LTR (long-term reference) have in average an approximately 1/2 lower intensity compared to other samples.

```

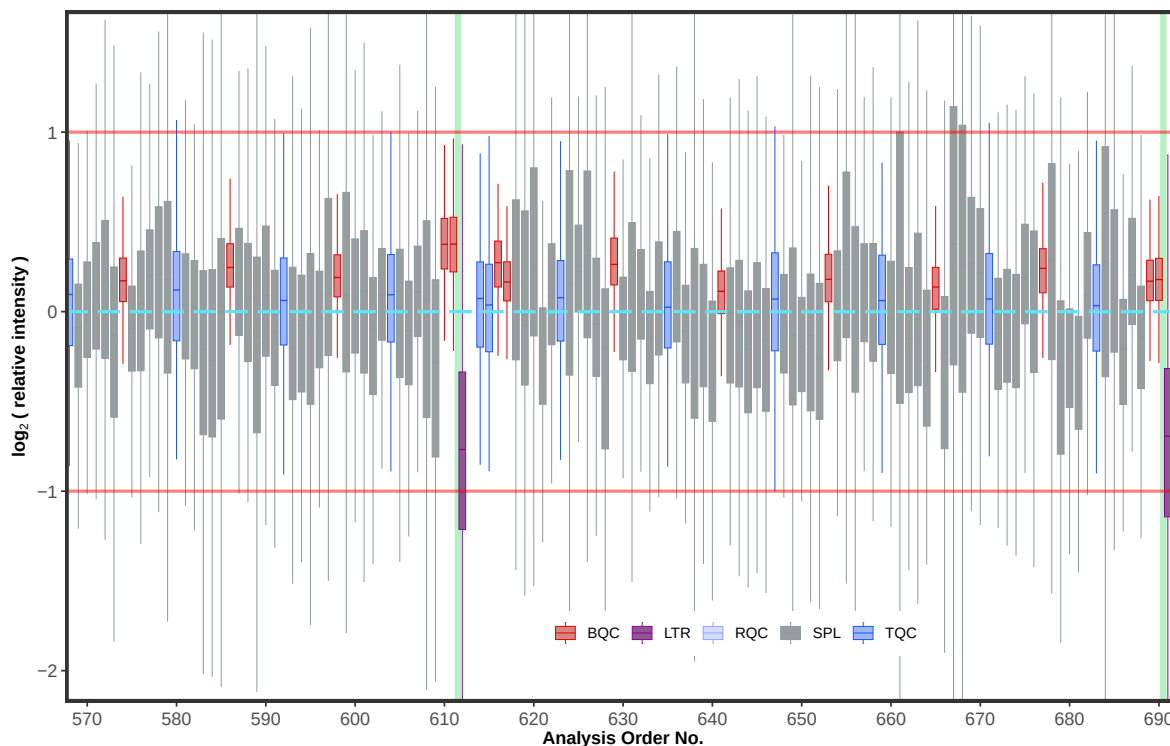
mexp_initial_temp <- mexp_initial
mexp_initial_temp <- exclude_analyses(
  mexp_initial_temp,
  analyses = "Longit_batch6_51",
  clear_existing = TRUE)
#> i 1 analyses were excluded for downstream processing. Please reprocess data.
res <- plot_rla_boxplot(
  data = mexp_initial_temp,
  rla_type_batch = c("across"),
  variable = "intensity",

```

```

qc_types = c("BQC", "SPL", "LTR", "TQC", "RQC"),
filter_data = FALSE,
exclude_feature_filter = "ISTD",
plot_range = c(570, 690), # get_batch_boundaries(mexp, batch_indices = 6),
y_lim = c(-2, 1.5),
include_istd = FALSE,
include_qualifier = FALSE,
min_feature_intensity = 500,
show_timestamp = FALSE,
outlier_method = "fold",
outlier_k = c(-1, 1),
outlier_exclude = FALSE,
x_gridlines = FALSE,
batch_zebra_stripe = FALSE,
base_font_size = 6,
show_plot = FALSE,
linewidth = 0.1
)
#> i No outliers found.
fig4b <- res$plot
fig4b

```



In the plot above we excluded the internal standards (ISTDs). In contrast below, we plot the distributions of the ISTDs only. We see that also the ISTDs have approximately 1/2 lower intensity in the LTR samples compared to the other samples.

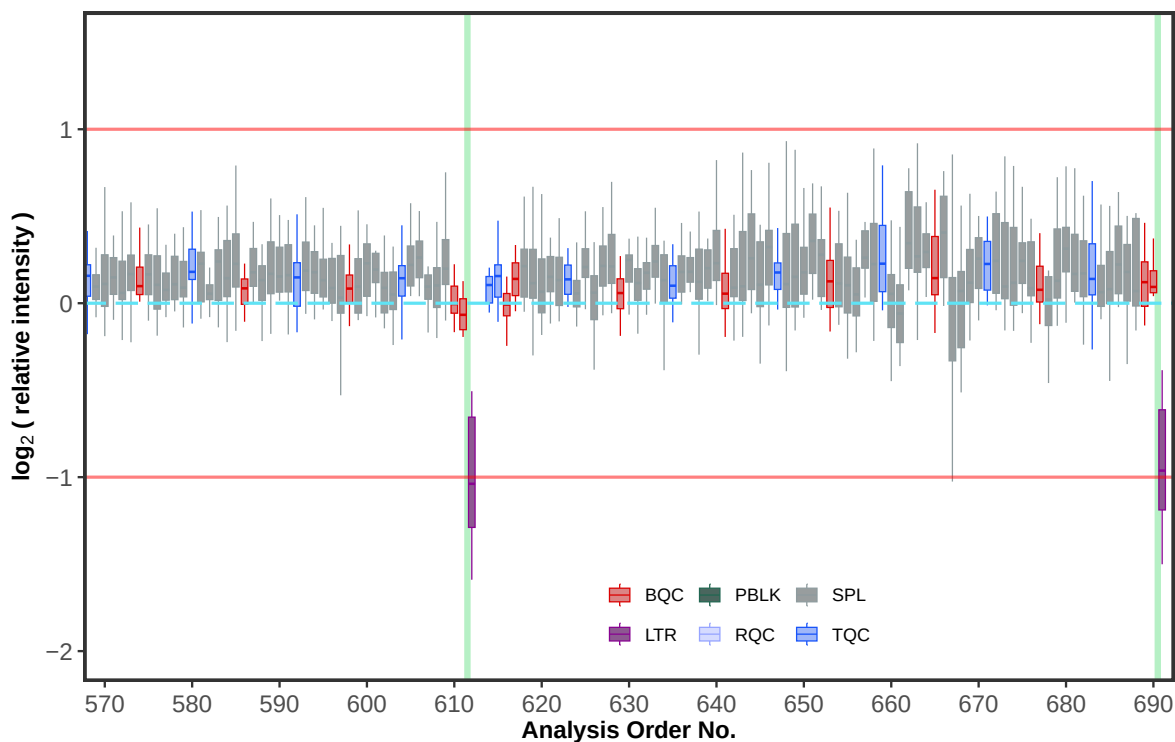
```
mexp_initial_temp2 <- mexp_initial
mexp_initial_temp2@dataset <- mexp_initial_temp2@dataset |> filter(is_istd)

extfig2b <- plot_rla_boxplot(
  data = mexp_initial_temp2,
  rla_type_batch = c("across"),
  variable = "intensity",
  qc_types = c("BQC", "SPL", "LTR", "TQC", "RQC", "PBLK"),
  filter_data = FALSE,
  include_feature_filter = "ISTD",
  plot_range = c(570, 690),
  y_lim = c(-2, 1.5),
  include_istd = TRUE,
  include_qualifier = FALSE,
  min_feature_intensity = 500,
  show_timestamp = FALSE,
  outlier_method = "fold",
```

```

outlier_k = c(-1, 1),
outlier_exclude = FALSE,
x_gridlines = FALSE,
batch_zebra_stripe = FALSE,
show_plot = FALSE,
base_font_size = 8,
linewidth = 0.2
)$plot
#> i Found 5 outliers in the 919 shown analyses
extfig2b

```



This may suggest that the measured volume was lower for the LTR samples, or that twice the amount of extraction solvent was added, both of which will lead to reduction of all analytes, including ISTDs.

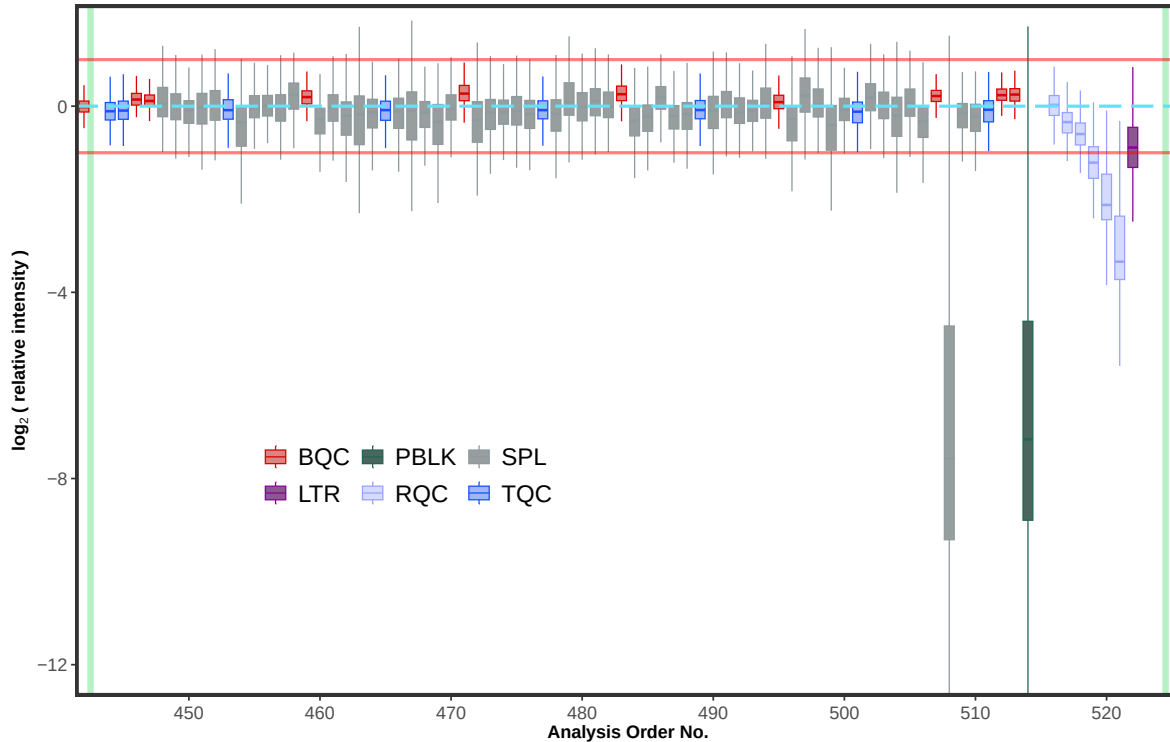
A batch-by-batch inspection revealed furthermore that in batch 6, the analysis/sample “Longit_batch6_51” at position 464 had significantly lower intensities of non-ISTD feature, similar to the intensity in Blanks.

```

mexp_initial_temp1 <- mexp_initial
mexp_initial_temp1@dataset <- mexp_initial_temp1@dataset |> filter(!is_istd)

extfig2c <- plot_rla_boxplot(
  data = mexp_initial_temp1,
  rla_type_batch = c("across"),
  variable = "intensity",
  qc_types = c("BQC", "SPL", "LTR", "TQC", "RQC", "PBLK"),
  filter_data = FALSE,
  plot_range = get_batch_boundaries(mexp_initial_temp2, batch_indices = 6),
  y_lim = c(-12, 1.5),
  include_istd = FALSE,
  #include_feature_filter = "ISTD",
  include_qualifier = FALSE,
  min_feature_intensity = 500,
  show_timestamp = FALSE,
  outlier_method = "fold",
  outlier_k = c(-1, 1),
  outlier_exclude = FALSE,
  x_gridlines = FALSE,
  batch_zebra_stripe = FALSE,
  base_font_size = 6,
  show_plot = FALSE,
  linewidth = 0.2)$plot +
  theme(
    strip.text = ggplot2::element_text(size = 8),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    axis.text = element_text(size = 8),
    #legend.direction = "vertical",
    legend.text = element_text(size = 8),
    legend.title = element_text(size = 8),
    legend.key.size = unit(8, "pt"),
    legend.position.inside = c(0.3, 0.32))
#> i Found 1 outliers in the 919 shown analyses
extfig2c

```

However, the ISTDs were present at comparable levels as other analyses/samples. This likely suggests that this study sample was not added to the extraction vial before extraction. The sample will be excluded from further analysis.

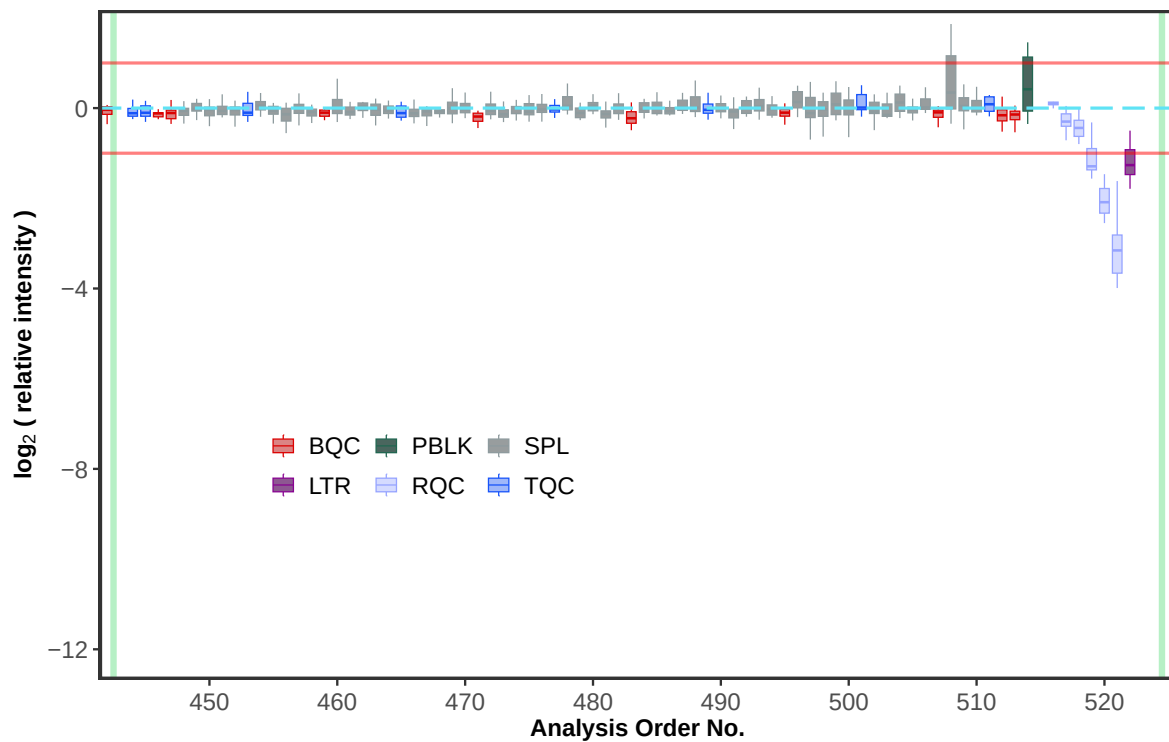
```
mexp_initial_temp2 <- mexp_initial
mexp_initial_temp2@dataset <- mexp_initial_temp2@dataset |> filter(is_istd)

extfig2d <- plot_rla_boxplot(
  data = mexp_initial_temp2,
  rla_type_batch = c("across"),
  variable = "intensity",
  qc_types = c("BQC", "SPL", "LTR", "TQC", "RQC", "PBLK"),
  filter_data = FALSE,
  plot_range = get_batch_boundaries(mexp_initial_temp2, batch_indices = 6),
  y_lim = c(-12, 1.5),
  include_istd = TRUE,
  #include_feature_filter = "ISTD",
  include_qualifier = FALSE,
  min_feature_intensity = 500,
  show_timestamp = FALSE,
  outlier_method = "fold",
```

```

outlier_k = c(-1, 1),
outlier_exclude = FALSE,
x_gridlines = FALSE,
batch_zebra_stripe = FALSE,
base_font_size = 8,
show_plot = FALSE,
linewidth = 0.2)$plot +
  theme(
    strip.text = ggplot2::element_text(size = 8),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    axis.text = element_text(size = 8),
    #legend.direction = "vertical",
    legend.text = element_text(size = 8),
    legend.title = element_text(size = 8),
    legend.key.size = unit(8, "pt"),
    legend.position.inside = c(0.3, 0.32))
#> i Found 5 outliers in the 919 shown analyses
extfig2d

```

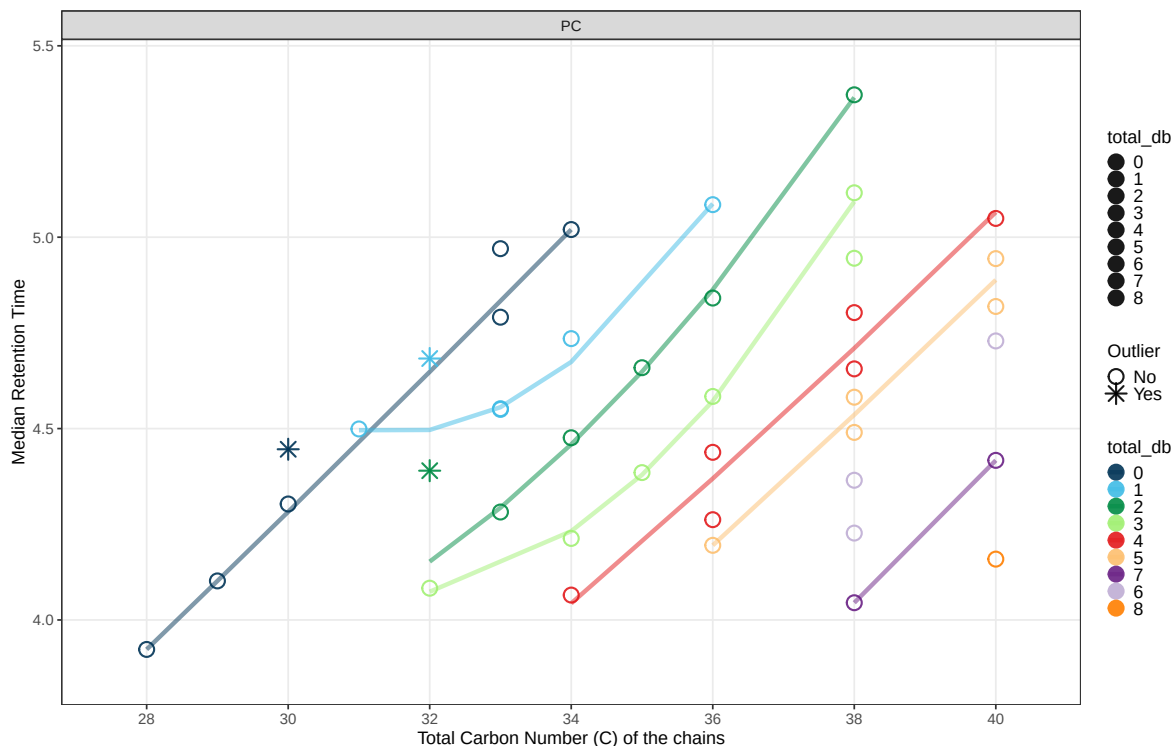


5. Peak picking QC

To check for potential peak picking errors, we plot the retention time (RT) of lipid features against the total carbon count and double bond number.

```
mexp_initial <- exclude_analyses(mexp_initial,
                                analyses = "Longit_batch6_51",
                                clear_existing = TRUE)
#> i 1 analyses were excluded for downstream processing. Please reprocess data.
mexp_temp <- mexp_initial
mexp_temp@dataset <- mexp_temp@dataset |> filter(str_detect(feature_id, "^PC \\d"))

fig4c <- plot_rt_vs_chain(
  mexp_temp,
  qc_types = "SPL",
  x_var = "total_c",
  base_font_size = 6) +
  theme(
    legend.position = "right",
    legend.direction = "vertical",
    legend.text = element_text(size = base_font_size*0.8),
    legend.title = element_text(size = base_font_size*0.8),
    legend.key.size = unit(base_font_size *0.8, "pt"),
    legend.position.inside = c(0.1, 0.7)
  )
#> i The following features were flagged as potential annotation outliers: PC 30:0 (b), PC 3
fig4c
```



We observe several potential peak picking errors, especially among the SM, which are likely caused by erroneous peak picking of the isobaric PC feature isotopologues.

6. Feature correlation analysis

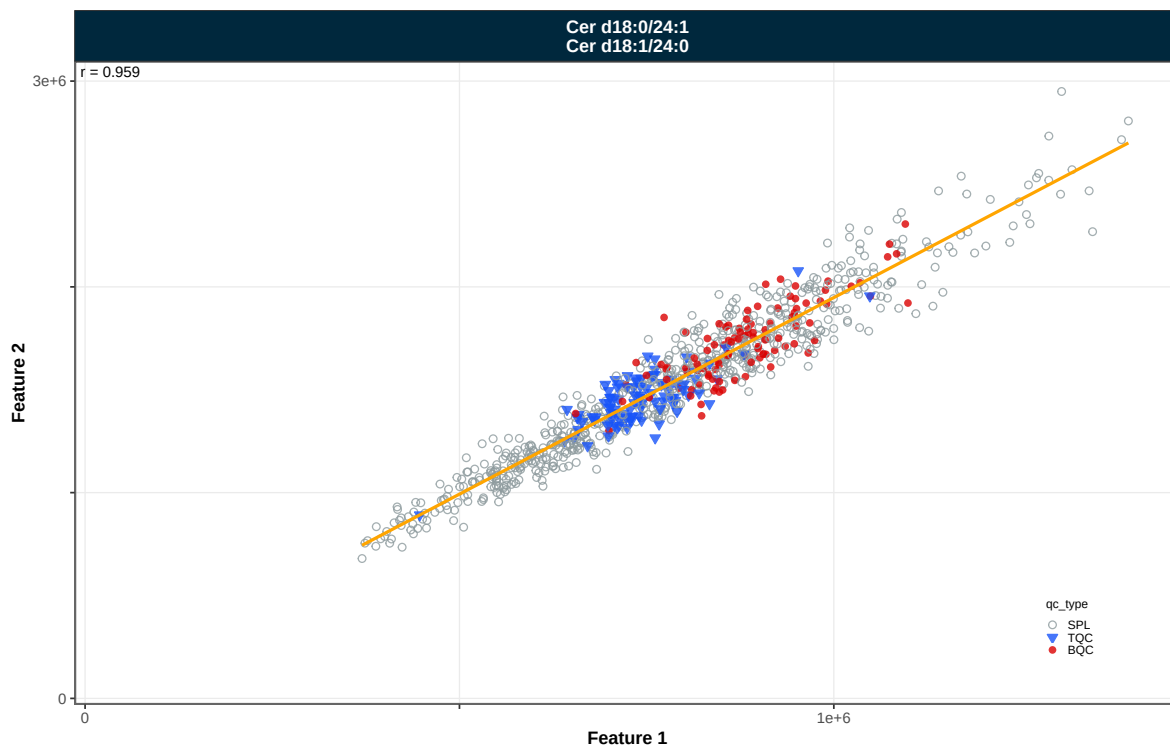
As another check for potential peak picking errors, highly correlating features are plotted.

```
# this below is to exclude a sample that has a very low intensity for all features, see next
# details
mexp_initial <- exclude_analyses(
  mexp_initial,
  analyses = "Longit_batch6_51",
  clear_existing = TRUE)
#> i 1 analyses were excluded for downstream processing. Please reprocess data.
fig4d <- plot_feature_correlations(
  mexp_initial,
  variable = "intensity",
  include_feature_filter = c(
    "Cer d18:1/24:0", "Cer d18:0/24:1"),
  qc_types = c("SPL", "BQC", "TQC"),
```

```

cor_min = 0.1, sort_by_corr = TRUE, return_plot = TRUE, show_progress = F,
log_scale = FALSE, cols_page = 1, rows_page = 1,
font_base_size = 6)[[1]] +
theme(
  strip.text = ggplot2::element_text(size = 6),
  #aspect.ratio = 0.9,
  legend.position = "inside",
  axis.text = element_text(size = 5),
  legend.direction = "vertical",
  legend.text = element_text(size = 6*0.7),
  legend.title = element_text(size = 6*0.7),
  legend.key.size = unit(6 * 0.7, "pt"),
  legend.position.inside = c(0.9, 0.12)
)
#> Generating plots (1 page)
fig4d

```



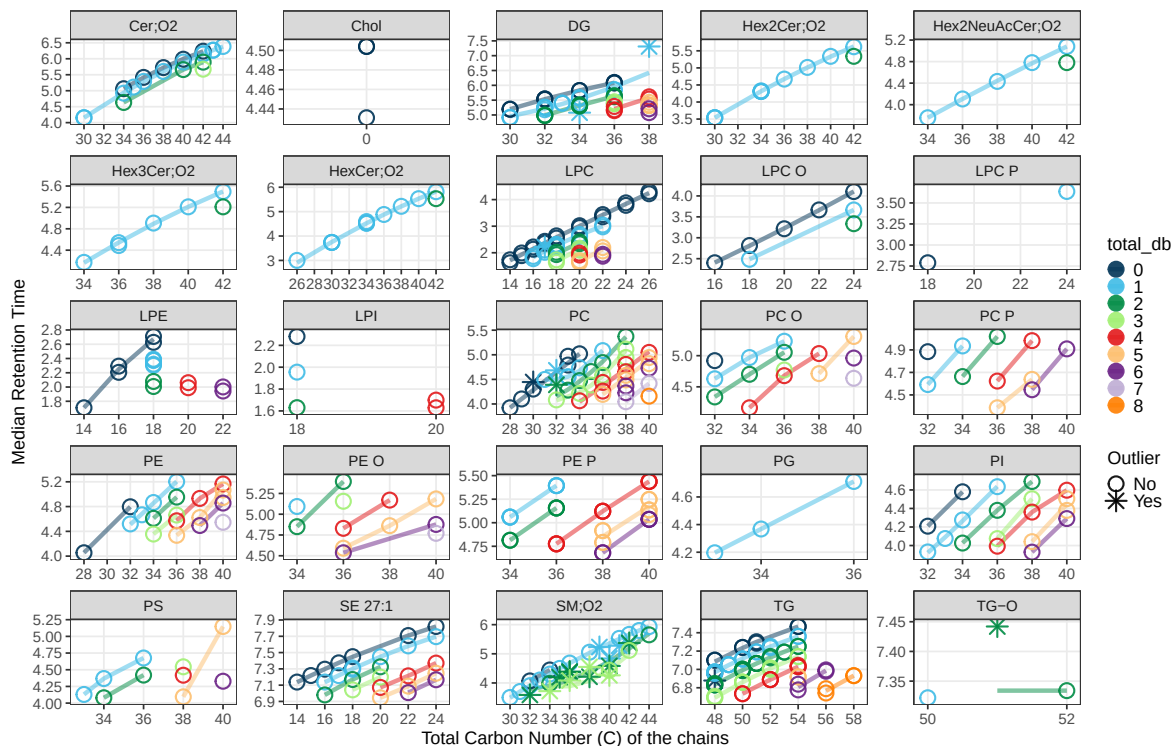
7. Fix peak integration and re-import data (V2)

We now went back to the MRMkit workflow and corrected the peak integration errors. The corrected data is available in the file `sPerfect_LongCirc_MRMkit.csv`. We import the corrected data, which we will use from now.

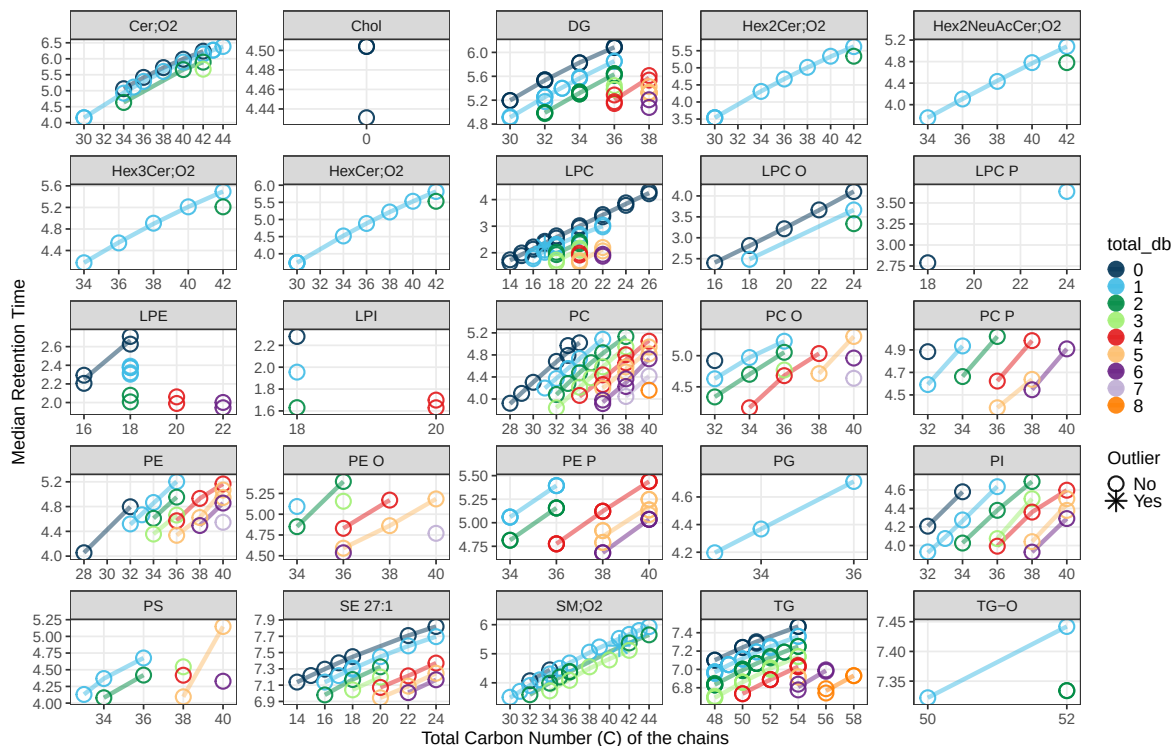
```
mexp <- MRMhubExperiment()
mexp <- import_data_mrmhub(
  mexp,
  "./data/dataset-1/data/sPerfect_LongCirc_MRMkit_Final.csv"
)
#> v Imported 937 analyses with 482 features
#> i `feature_area` selected as default feature intensity. Modify with `set_intensity_var()`
#> v Analysis metadata associated with 937 analyses.
#> v Feature metadata associated with 482 features.
```

We also directly plot the RT against chain lengths again, which confirms that the peak picking errors seems to have been corrected.

```
extfig3a <- plot_rt_vs_chain(mexp_initial, qc_types = "SPL", x_var = "total_c", base_font_size = 12,
  theme(
    legend.position = "right",
    legend.direction = "vertical",
    legend.text = element_text(size = base_font_size*0.8),
    legend.title = element_text(size = base_font_size*0.8),
    legend.key.size = unit(base_font_size * 0.8, "pt"),
    legend.position.inside = c(0.1, 0.7))
#> i The following features were flagged as potential annotation outliers: DG 18:1_20:0 [-18]
extfig3a
```



```
extfig3b <- plot_rt_vs_chain(mexp, qc_types = "SPL", x_var = "total_c", base_font_size = 6)+
  theme(
    legend.position = "right",
    legend.direction = "vertical",
    legend.text = element_text(size = base_font_size*0.8),
    legend.title = element_text(size = base_font_size*0.8),
    legend.key.size = unit(base_font_size *0.8, "pt"),
    legend.position.inside = c(0.1, 0.7))
#> i The following features were flagged as potential annotation outliers: LPC 17:1 (c), SM 4
extfig3b
```



8. PCA to check for potential technical outliers

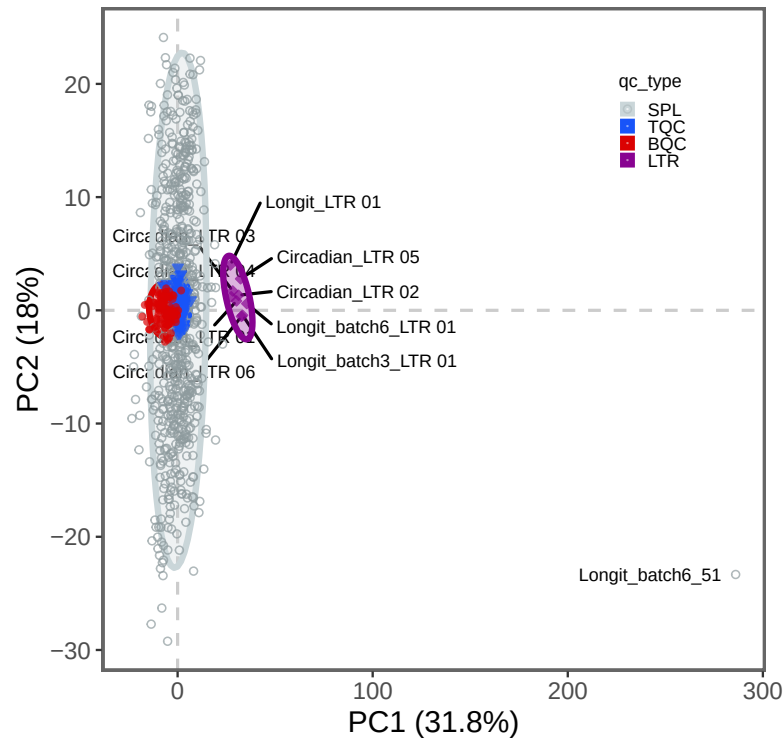
```
extfig4a <- plot_pca(
  data = mexp,
  variable = "intensity",
  filter_data = FALSE,
  pca_dims = c(1,2),
  labels_threshold_mad = 4,
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  ellipse_variable = "qc_type",
  log_transform = TRUE,
  point_size = 1, point_alpha = 0.7, font_base_size = 8, ellipse_alpha = 0.3,
  include_istd = FALSE, show_labels = TRUE, label_font_size = 2,
  shared_labeltext_hide = NA) + theme(
  plot.title = element_blank(),
  aspect.ratio = 1,
  legend.position = "inside",
  legend.direction = "vertical",
  legend.text = element_text(size = 9*0.7),
```



```

legend.title = element_text(size = 9*0.7),
legend.key.size = unit(7 *0.7, "pt"),
legend.position.inside = c(0.83, 0.83))
#> ! 9 features contained missing or non-numeric values and were excluded.
#> i The PCA was calculated based on `feature_intensity` values of 465 features.
extfig4a

```



9. Exclude outlier analysis/sample and replot PCA

```

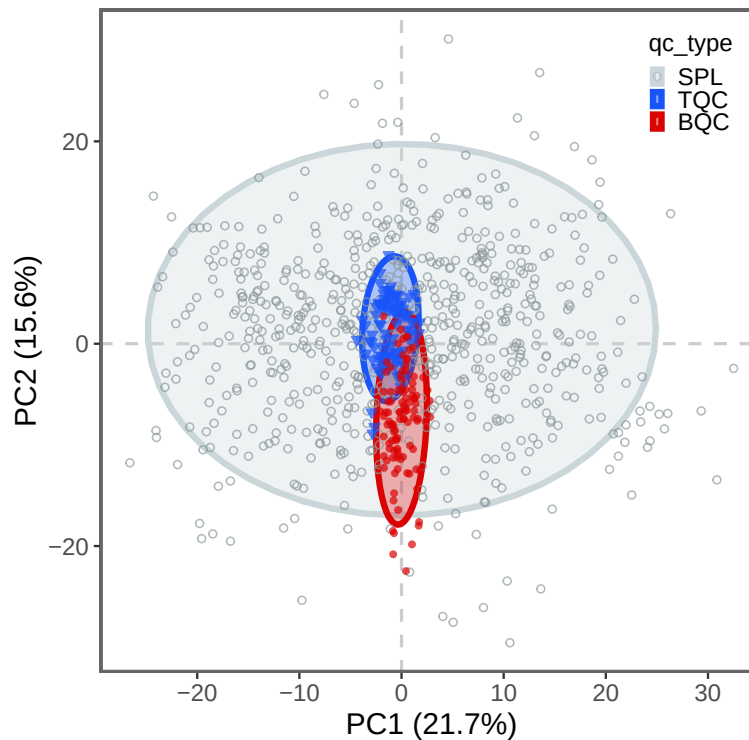
mexp <- exclude_analyses(
  mexp,
  analyses = "Longit_batch6_51", clear_existing = FALSE)
#> i A total of 1 analyses are now excluded for downstream processing. Please reprocess data
fig4e <- plot_pca(
  data = mexp,
  variable = "intensity",
  filter_data = FALSE,
  pca_dims = c(1,2),

```

```

labels_threshold_mad = 4,
qc_types = c("SPL", "BQC", "TQC"),
ellipse_variable = "qc_type",
log_transform = TRUE,
point_size = 1, point_alpha = 0.7, font_base_size = 8, ellipse_alpha = 0.3,
include_istd = FALSE, show_labels = FALSE, label_font_size = 2,
shared_labeltext_hide = NA) +
  theme(
    plot.title = element_blank(),
    legend.position = "inside",
    legend.direction = "vertical",
    legend.text = element_text(size = 12*0.7),
    legend.title = element_text(size = 12*0.7),
    legend.key.size = unit(8 *0.7, "pt"),
    legend.position.inside = c(0.89, 0.89))
#> ! 9 features contained missing or non-numeric values and were excluded.
#> i The PCA was calculated based on `feature_intensity` values of 465 features.
fig4e

```



10. Importing detailed metadata

To proceed with further processing, we require additional metadata describing the analyses/samples, features, internal standards (ISTDs), QC samples, and response curves. We are using the MRMhub MSorganizer template, an Excel template that provides a user-friendly solution to collect, organise and pre-validate analysis metadata. However, other formats to import metadata are available as well, such as from CSV files, individual Excel tables, or R data frames.

After metadata import, a summary of the validation of data/metadata integrity is returned. For example, duplicate or missing data, such as IDs will be highlighted. The user can choose to ignore warnings (`ignore_warnings = TRUE`) and proceed with the analysis. Errors however, such as missing IDs or duplicate IDs must always be fixed before analysis can proceed.

```
file_path <- "../data/dataset-1/data/sPerfect_LongCirc_Metadata.xlsm"
mexp <- import_metadata_msorganiser(mexp, file_path, ignore_warnings = TRUE)
#> ! Metadata has following warnings and notifications:
#> -----
#>   Type Table   Column      Issue                                Count
#> 1 W*   Features feature_id Feature(s) not in analysis data         28
#> 2 W*   Features feature_id Feature(s) without metadata              5
#> -----
#> E = Error, W = Warning, W* = Supressed Warning, N = Note
#> -----
#> v Analysis metadata associated with 937 analyses.
#> v Feature metadata associated with 477 features.
#> i 4 invalid features (as defined in the metadata) were excluded
#> v Internal Standard metadata associated with 18 ISTDs.
#> v Response curve metadata associated with 24 annotated analyses.
```

11. RunScatter plots

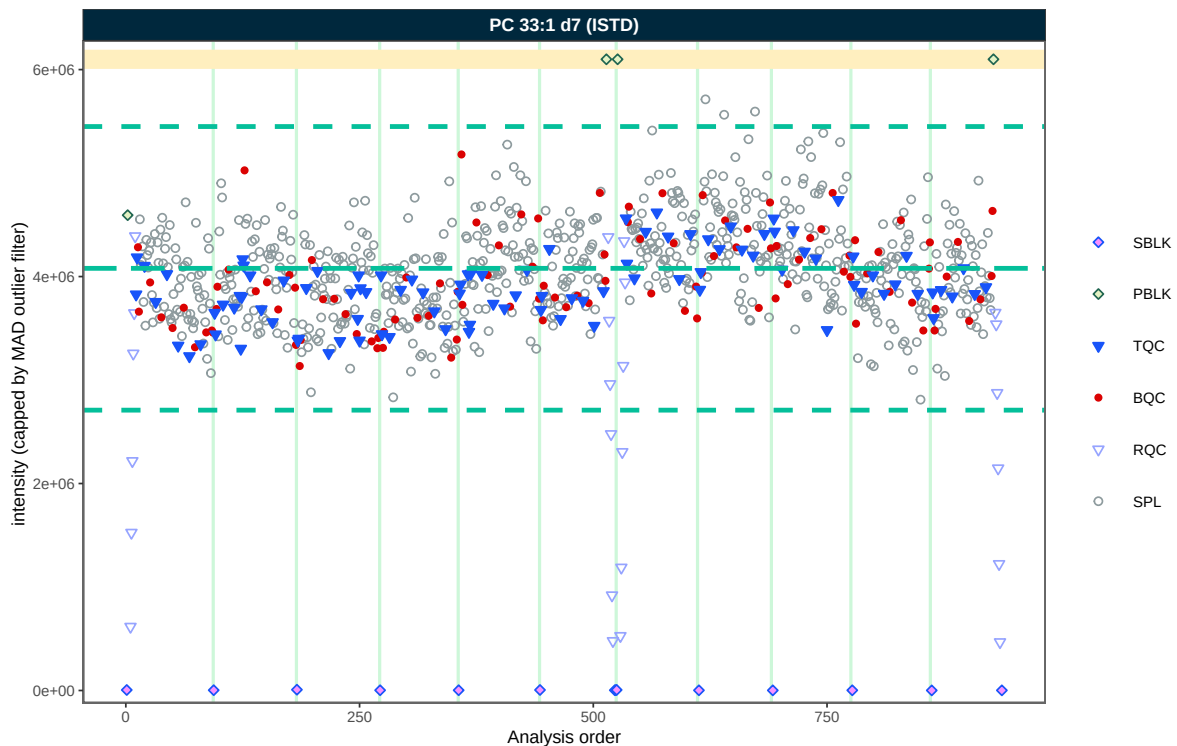
We now plot the peak areas (`intensity`) of all ISTD features across all analyses/samples.

```
fig4f <- plot_runscatter(mexp, variable = "intensity",
  #include_feature_filter = "ISTD",
  qc_types = c("SPL", "BQC", "TQC", "PBLK", "RQC", "SBLK"),
  include_feature_filter = example_species_istd,
  show_reference_lines = TRUE,
  ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
```

```

    point_size = 1,
    base_font_size = 6,
    cols_page = 1,
    rows_page = 1,
    cap_outliers = TRUE,
    reference_sd_shade = FALSE,
    #batch_zebra_stripe = TRUE,
    output_pdf = FALSE,
    show_progress = FALSE,
    path = "rt_all.pdf",
    return_plots = TRUE)[[1]]+
theme(
  legend.position = "inside",
  legend.direction = "horizontal",
  legend.text = element_text(size = base_font_size*0.8),
  legend.title = element_text(size = base_font_size*0.8),
  legend.key.size = unit(base_font_size *0.8, "pt"),
  legend.position.inside = c(0.7, 0.1))
#> Generating plots (1 page)

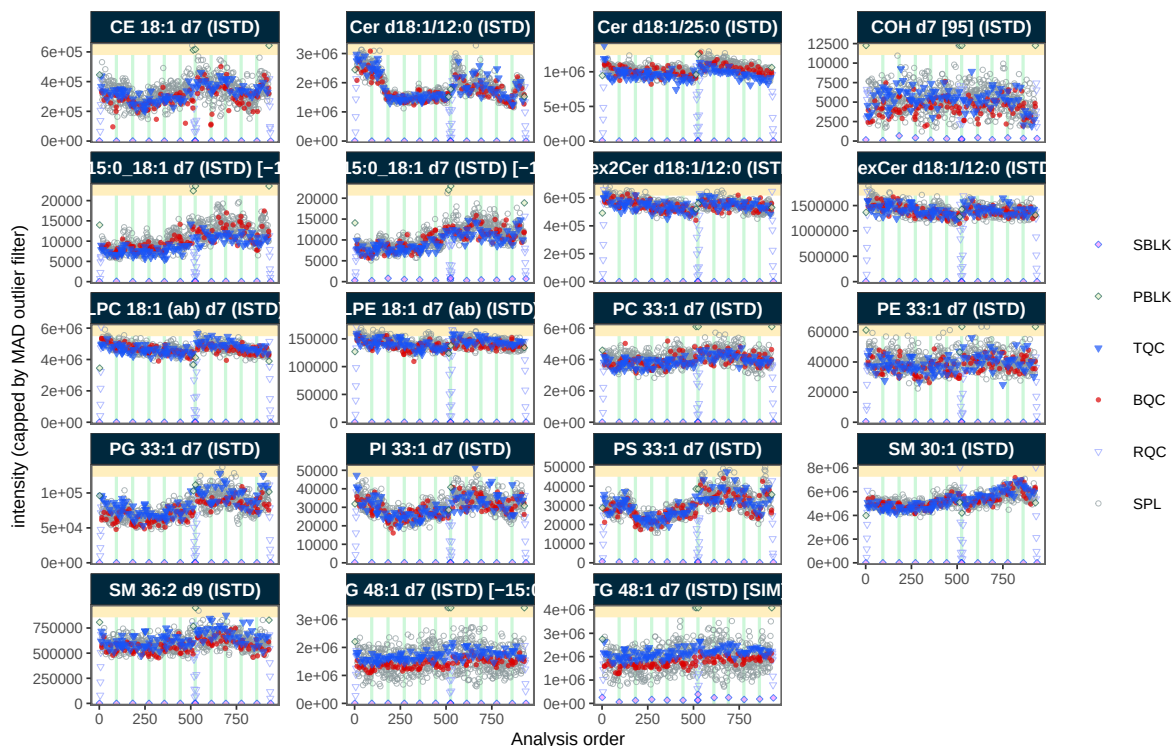
```



Additionally, we plot all ISTD features in a multi-page PDF to check for potential issues with

individual ISTDs.

```
extfig5 <- plot_runscatter(mexp, variable = "intensity",
  include_feature_filter = "ISTD", include_qualifier = F,
  qc_types = c("SPL", "BQC", "TQC", "PBLK", "RQC", "SBLK"),
  show_reference_lines = FALSE, ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
  point_size = .7,
  point_border_width = 0.2,
  point_transparency = .7,
  base_font_size = 6,
  cols_page = 4,
  rows_page = 5,
  cap_outliers = TRUE,
  reference_sd_shade = FALSE,
  #batch_zebra_stripe = TRUE,
  show_progress = FALSE,
  output_pdf = FALSE,
  path = "output/area_istd_all.pdf", return_plots = TRUE)
#> Generating plots (1 page)
```

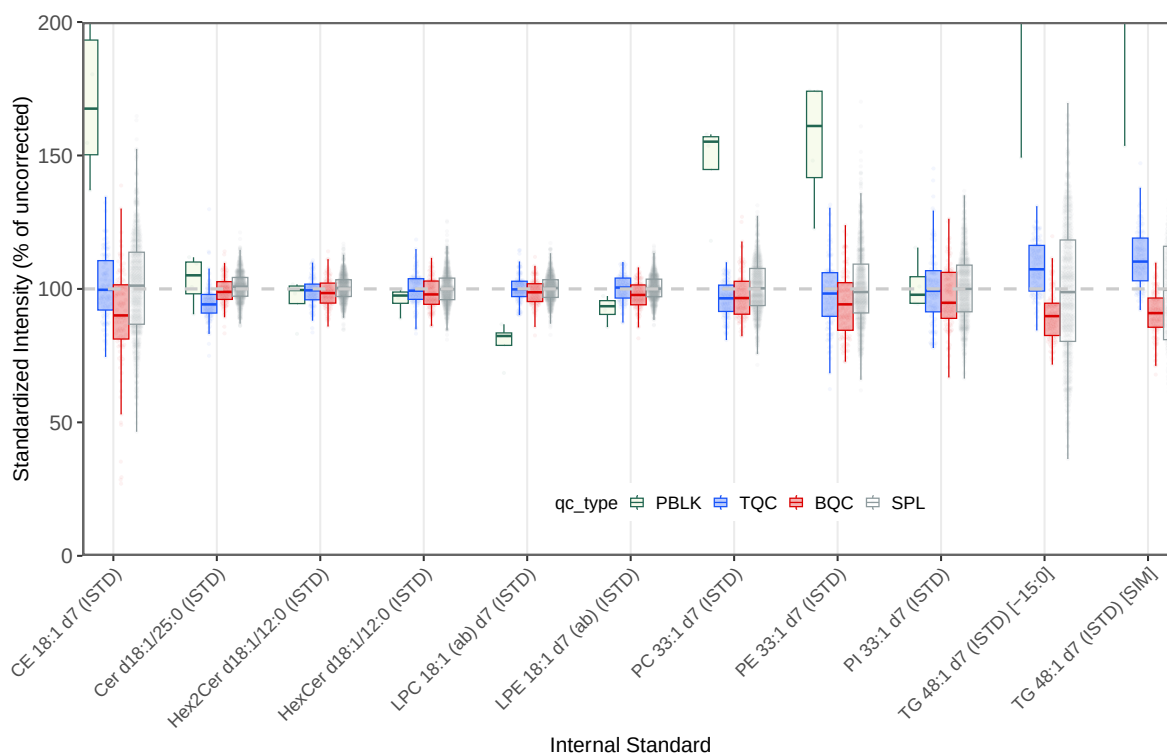


12. Matrix effects

To check for potential matrix effects, we plot the peak areas (**intensity**) of all ISTD features in different QC types.

```
mexp_temp <- mexp
mexp_temp@dataset <- mexp_temp@dataset |> filter(!str_detect(feature_id, "TG 48:0"))

fig4g <- plot_qc_matrixeffects(
  mexp_temp,
  variable = "intensity",
  y_lim = c(0,200),
  point_alpha = 0.05,
  box_alpha = 0.3,
  point_size = 0.2, box_linewidth = 0.2,
  font_base_size = 7,
  include_feature_filter = "P[CIE] | TG | Cer | CE",
  min_median_value = 3000)
fig4g
```



13. Normalization and Quantification

```
mexp <- normalize_by_istd(mexp)
#> ! Interfering features defined in metadata, but no correction was applied. Use `correct_in
#> v 435 features normalized with 17 ISTDs in 937 analyses.
mexp <- quantify_by_istd(mexp)
#> v 449 feature concentrations calculated based on 28 ISTDs and sample amounts of 937 analy
#> i Concentrations are given in mol/L.
```

14. Drift and Batch Correction

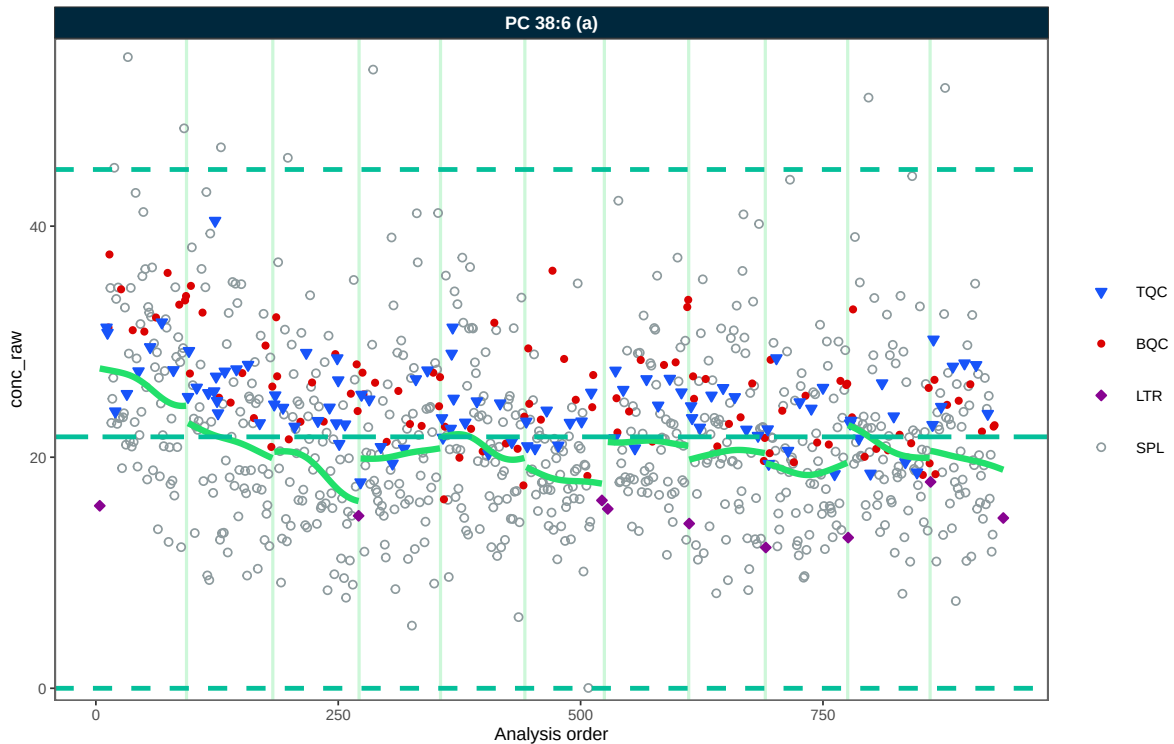
```
mexp <- correct_drift_gaussiankernel(
  mexp,
  variable = "conc",
  ref_qc_types = "SPL",
  batch_wise = TRUE,
  kernel_size = 20,
  outlier_filter = TRUE,
  outlier_ksd = 3,
  recalc_trend_after = TRUE,
  show_progress = FALSE)
#> i Applying `conc` drift correction...
#> i 6 feature(s) contain one or more zero or negative `conc` values. Verify your data or use
#> ! 3 features showed no variation in the study sample's original values across analyses.
#> ! 3 features have invalid values after smoothing. NA will be returned for all values of
#> v Drift correction was applied to 446 of 449 features (batch-wise).
#> i The median CV change of all features in study samples was -0.52% (range: -7.93% to 1.56%)
mexp <- correct_batch_centering(
  mexp,
  ref_qc_types = "SPL",
  variable = "conc",)
#> i Adding batch correction on top of `conc` drift-correction.
#> v Batch median-centering of 11 batches was applied to drift-corrected concentrations of a
#> i The median CV change of all features in study samples was -0.97% (range: -30.20% to 50.8%)
```

15. Drift and Batch Correction


```

fig5c <- plot_runscatter(mexp, variable = "conc_raw",
  #include_feature_filter = "ISTD",
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  include_feature_filter = example_species,
  #y_min = 0.00,
  #y_max = 0.15,
  #plot_range = c(0, 910),
  show_reference_lines = TRUE, ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
  show_trend = TRUE,
  point_size = 1,
  base_font_size = 6,
  cols_page = 1,
  rows_page = 1,
  cap_outliers = FALSE,
  reference_sd_shade = FALSE,
  #batch_zebra_stripe = TRUE,
  show_progress = FALSE,
  output_pdf = FALSE,
  path = "rt_all.pdf", return_plot = TRUE)[[1]]+
  theme(
    legend.position = "inside",
    legend.direction = "horizontal",
    legend.text = element_text(size = base_font_size*0.8),
    legend.title = element_text(size = base_font_size*0.8),
    legend.key.size = unit(base_font_size *0.8, "pt"),
    legend.position.inside = c(0.7, 0.1))
#> Generating plots (1 page)

```



```
fig5d <- plot_runscatter(mexp, variable = "conc",
  #include_feature_filter = "ISTD",
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  include_feature_filter = example_species,
  #y_min = 0.00,
  #y_max = 0.9,
  #plot_range = c(0, 910),
  show_reference_lines = TRUE,
  ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
  show_trend = TRUE,
  point_size = 1,
  base_font_size = 6,
  cols_page = 1,
  rows_page = 1,
  cap_outliers = FALSE,
  reference_sd_shade = FALSE,
  #batch_zebra_stripe = TRUE,
  show_progress = FALSE,
```

```

        output_pdf = FALSE,
        path = "rt_all.pdf", return_plot = TRUE)[[1]]+
theme(
  legend.position = "inside",
  legend.direction = "horizontal",
  legend.text = element_text(size = base_font_size*0.8),
  legend.title = element_text(size = base_font_size*0.8),
  legend.key.size = unit(base_font_size *0.8, "pt"),
  legend.position.inside = c(0.7, 0.1))
#> Generating plots (1 page)

```

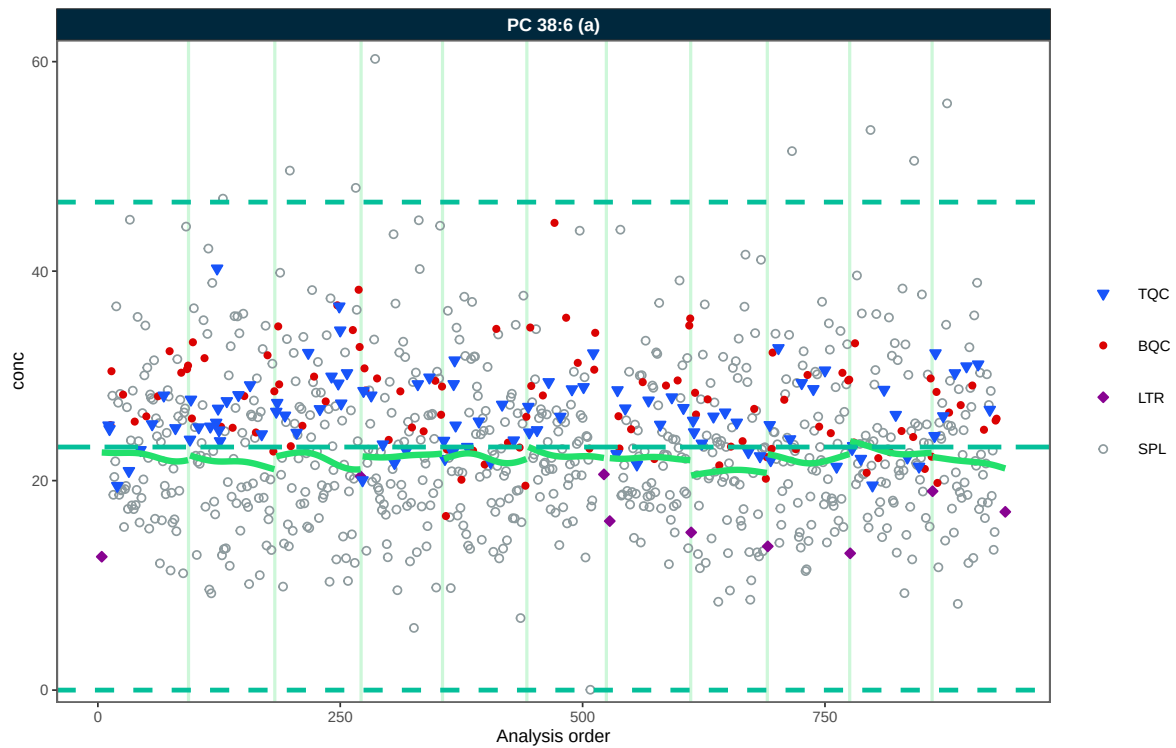
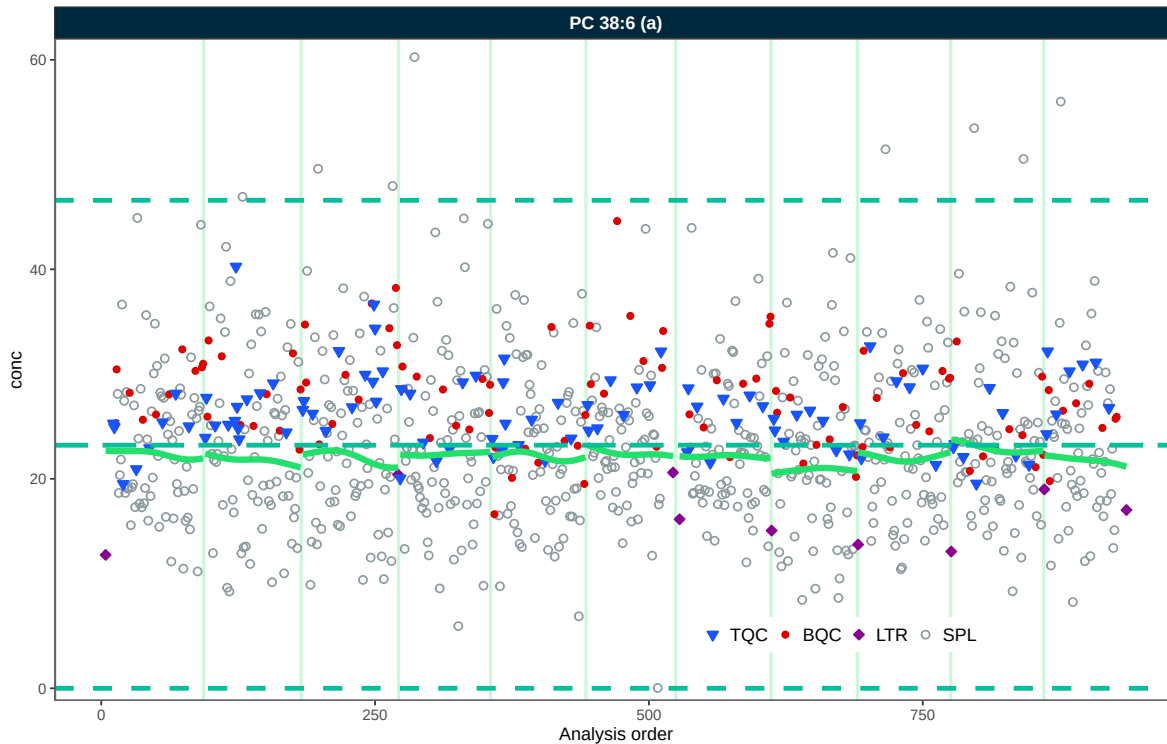


fig5d



16. Normalization and Correction QC

```
mexp_t1 <- calc_qc_metrics(mexp, use_robust_cv = TRUE, use_batch_medians = FALSE)
mexp_t1@metrics_qc <- mexp_t1@metrics_qc |>
  dplyr::filter(str_detect(feature_class, "^PC$|TG$"))

mexp_t2 <- calc_qc_metrics(mexp)
mexp_t2@metrics_qc <- mexp_t2@metrics_qc |>
  dplyr::filter(str_detect(feature_class, "^PC$|TG$"))

fig5a <- plot_normalization_qc(
  plot_type = "diff",
  data = mexp_t1,
  before_norm_var = "intensity",
  after_norm_var = "norm_intensity",
  y_lim = c(-15,20),
  x_lim = c(0,75),
  qc_types = c("TQC", "BQC", "SPL"),
  cols_page = 3,
```

```

font_base_size = 5,
point_size = 1,
facet_by_class = TRUE,
include_qualifier = FALSE) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    legend.position.inside = c(0.77, 0.27),
    axis.text = element_text(size = 4),
    #axis.title = element_blank(),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_text(size = 6*0.7),
    legend.key.size = unit(6 *0.7, "pt"))

fig5b <- plot_normalization_qc(
  data = mexp_t2,
  plot_type = "diff",
  before_norm_var = "norm_intensity",
  after_norm_var = "conc",
  y_lim = c(-15,20),
  x_lim = c(0,75),
  qc_types = c("TQC", "BQC", "SPL"),
  cols_page = 2,
  facet_by_class = TRUE,
  font_base_size = 5,
  point_size = 1,
  include_qualifier = FALSE) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    legend.position.inside = c(0.77, 0.27),
    axis.text = element_text(size = 4),
    #axis.title = element_blank(),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_text(size = 6*0.7),
    legend.key.size = unit(6 *0.7, "pt"))
fig5a

```

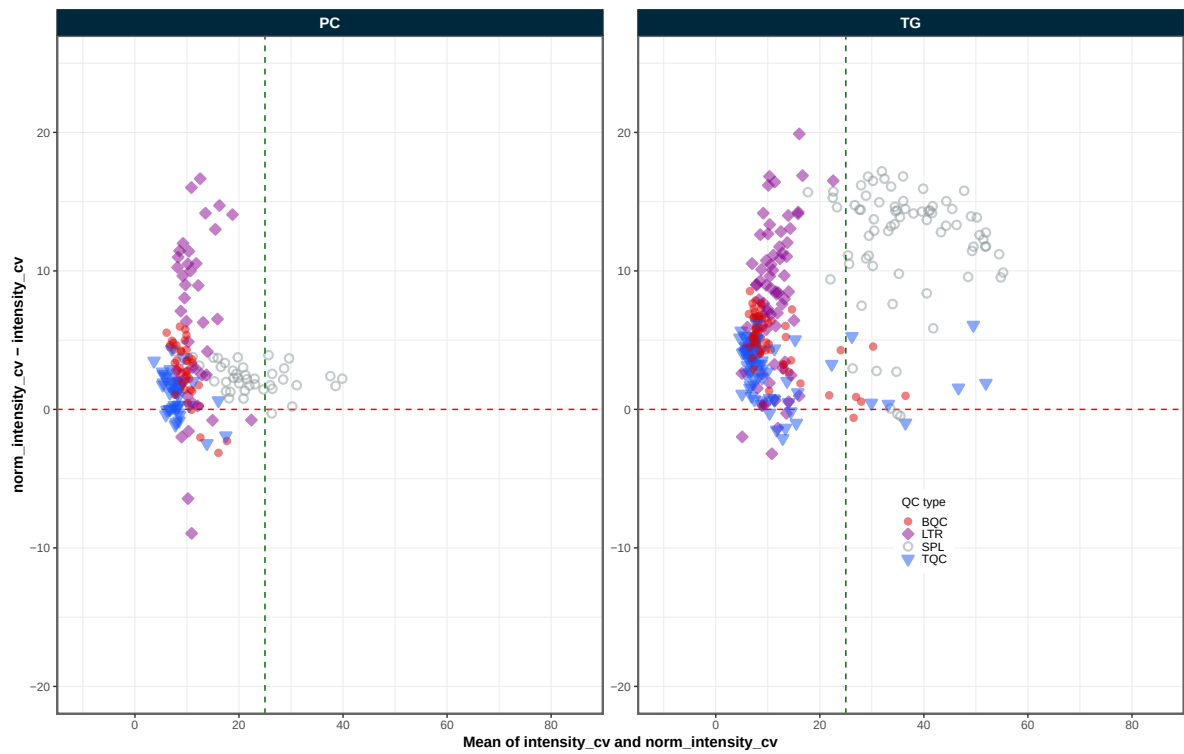
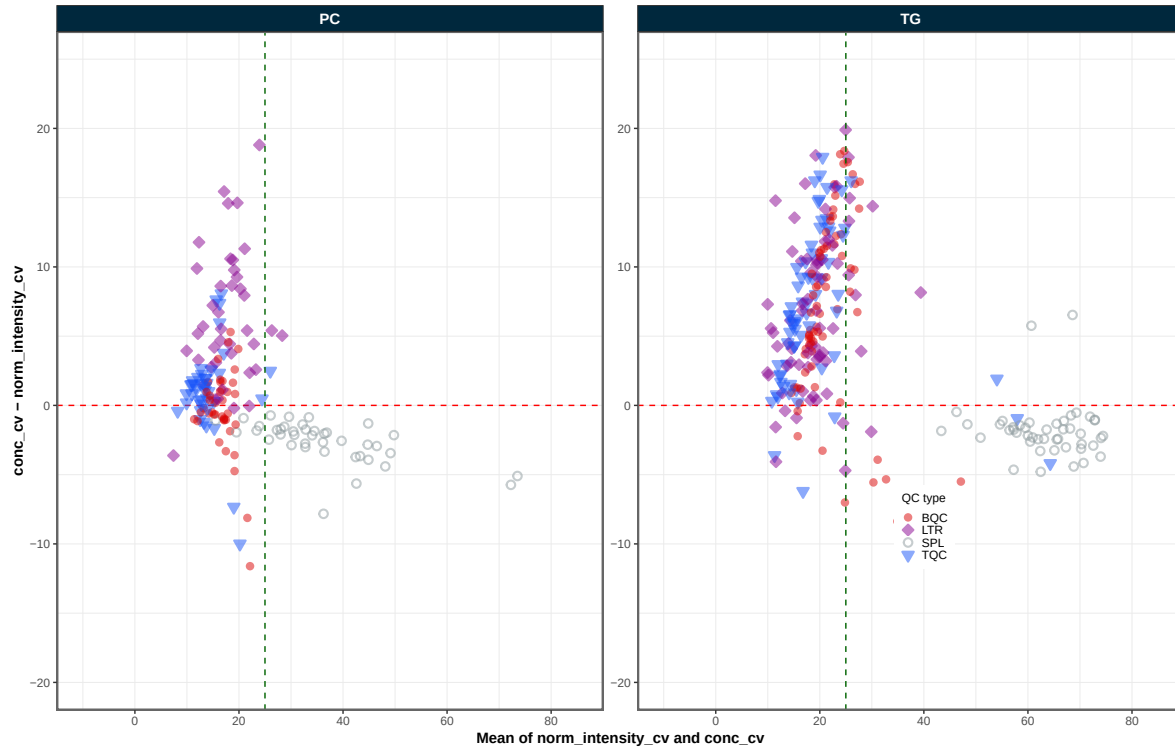


fig5b



```
mexp_temp <- calc_qc_metrics(mexp, use_robust_cv = TRUE, use_batch_medians = FALSE)

mexp_temp@metrics_qc <- mexp_temp@metrics_qc |>
  dplyr::filter(!str_detect(feature_id, "^TG 0|COH|PG"))

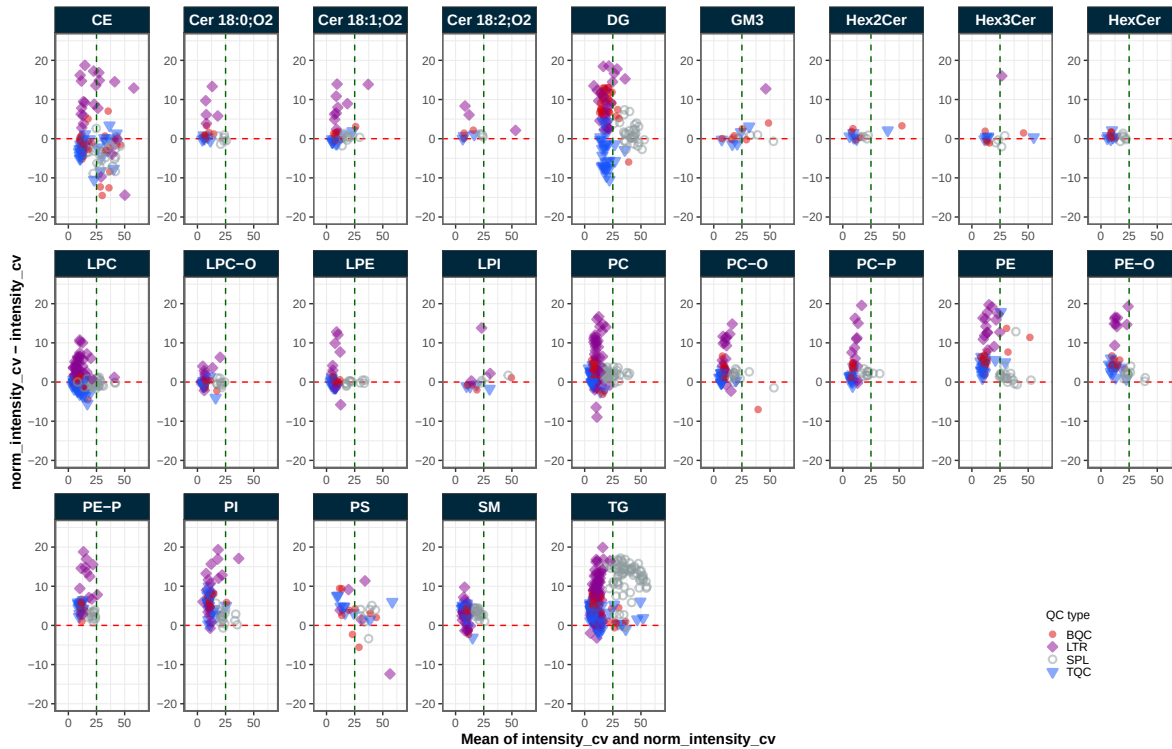
extfig6a <- plot_normalization_qc(
  data = mexp_temp,
  before_norm_var = "intensity",
  after_norm_var = "norm_intensity",
  plot_type = "diff",
  y_lim = c(-15, 20),
  x_lim = c(0, 60),
  qc_types = c("TQC", "BQC", "SPL"),
  cols_page = 9,
  font_base_size = 5,
  point_size = 0.85,
  facet_by_class = TRUE,
  include_qualifier = FALSE) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
```

```

legend.position = "inside",
legend.position.inside = c(0.9, 0.10),
axis.text = element_text(size = 4),
legend.direction = "vertical",
legend.text = element_text(size = 6*0.7),
legend.title = element_text(size = 6*0.7),
legend.key.size = unit(6 *0.7, "pt")

```

extfig6a



```

extfig6b <- plot_normalization_qc(
  data = mexp_temp,
  before_norm_var = "norm_intensity",
  after_norm_var = "conc",
  plot_type = "diff",
  y_lim = c(-15,20),
  x_lim = c(0,60),
  qc_types = c("TQC", "BQC", "SPL"),
  cols_page = 9,
  font_base_size = 5,
  point_size = 0.85,

```

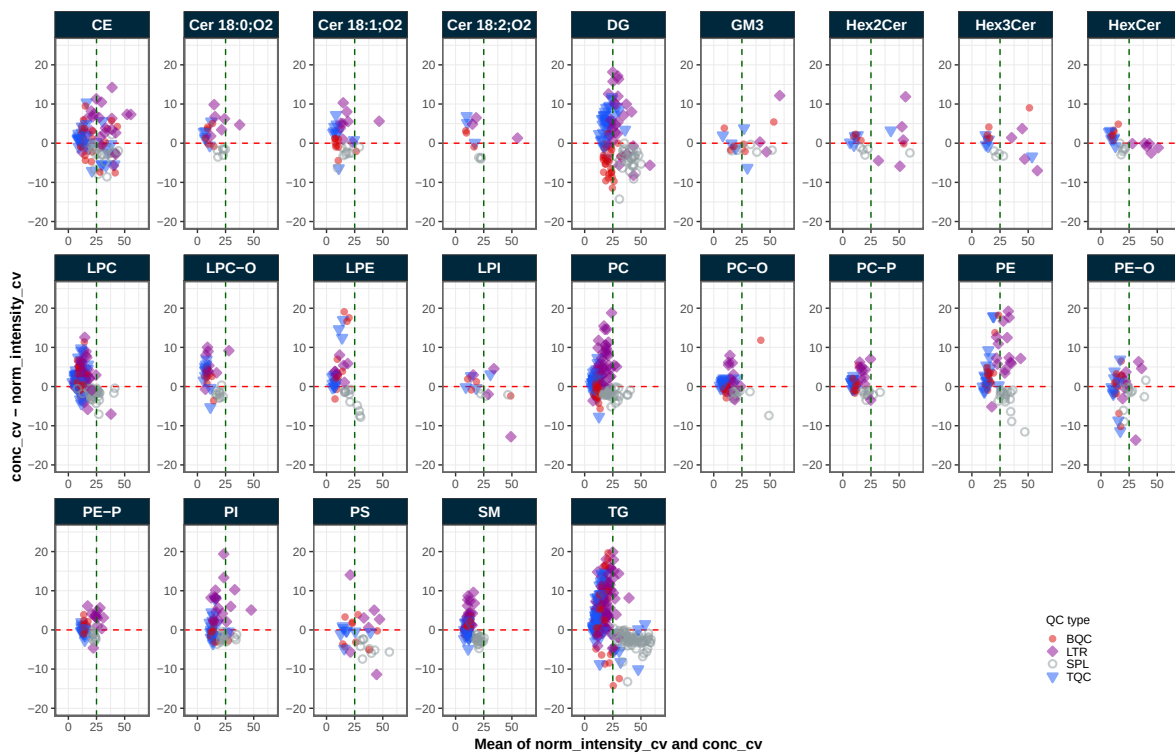


```

facet_by_class = TRUE,
include_qualifier = FALSE) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    legend.position.inside = c(0.9, 0.10),
    axis.text = element_text(size = 4),
    #axis.title = element_blank(),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_text(size = 6*0.7),
    legend.key.size = unit(6 *0.7, "pt"))

```

extfig6b



```

mexp_spline <- correct_drift_cubicspline(
  mexp,
  variable = "conc",
  ref_qc_types = "BQC",
  batch_wise = TRUE,

```

```

lambda = 0.5,
cv = TRUE,
recalc_trend_after = TRUE, replace_previous = TRUE)
#> i Replacing previous `conc` drift and batch corrections...
#> i 6 feature(s) contain one or more zero or negative `conc` values. Verify your data or use
#> ! 3 features showed no variation in the study sample's original values across analyses.
#> ! 4 features have invalid values after smoothing. NA will be returned for all values of
#> ! Smoothing failed for 4 feature(s) in all batches. Please check data, metadata, and fit p
#> ! Smoothing failed for 6 feature(s) in at least one batch: CE 15:0, COH [161], LPI 18:0, L
#> v Drift correction was applied to 445 of 449 features (batch-wise).
#> i The median CV change of all features in study samples was 0.08% (range: -3.49% to 14.38%)
mexp_spline <- correct_batch_centering(
  mexp_spline,
  ref_qc_types = "BQC",
  variable = "conc")
#> i Adding batch correction on top of `conc` drift-correction.
#> v Batch median-centering of 11 batches was applied to drift-corrected concentrations of a
#> i The median CV change of all features in study samples was 0.20% (range: -28.20% to 121.0%)
mexp_spline <- calc_qc_metrics(
  mexp_spline,
  use_robust_cv = TRUE,
  use_batch_medians = FALSE)

mexp_spline@metrics_qc <- mexp_spline@metrics_qc |>
  dplyr::filter(!str_detect(feature_id, "^TG 0|COH|PG"))

extfig6c <- plot_normalization_qc(
  data = mexp_spline, plot_type = "diff",
  before_norm_var = "norm_intensity",
  after_norm_var = "conc",
  y_lim = c(-20, 20),
  x_lim = c(0, 60),
  qc_types = c("TQC", "BQC", "SPL"),
  cols_page = 9,
  font_base_size = 5, cv_threshold_value = 25,
  point_size = 0.85,
  facet_by_class = TRUE,
  include_qualifier = FALSE) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",

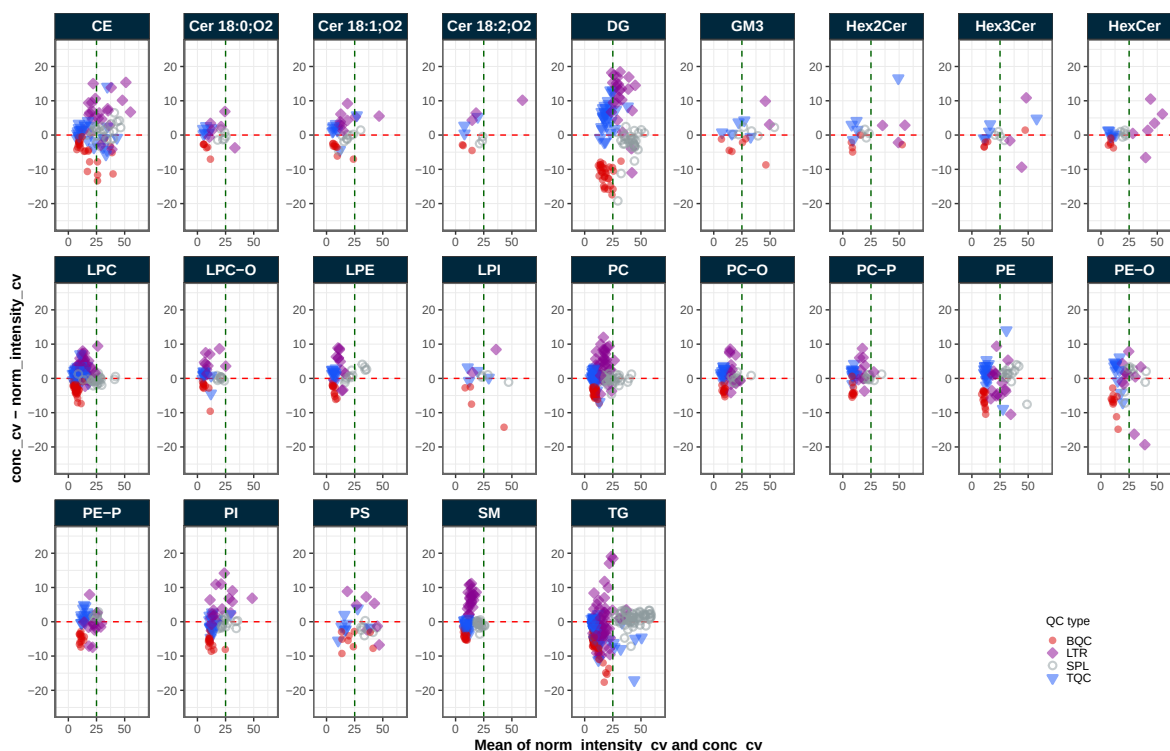
```

```

legend.position.inside = c(0.9, 0.10),
axis.text = element_text(size = 4),
#axis.title = element_blank(),
legend.direction = "vertical",
legend.text = element_text(size = 6*0.7),
legend.title = element_text(size = 6*0.7),
legend.key.size = unit(6 *0.7, "pt")

```

extfig6c



17. BQC vs TQC variability

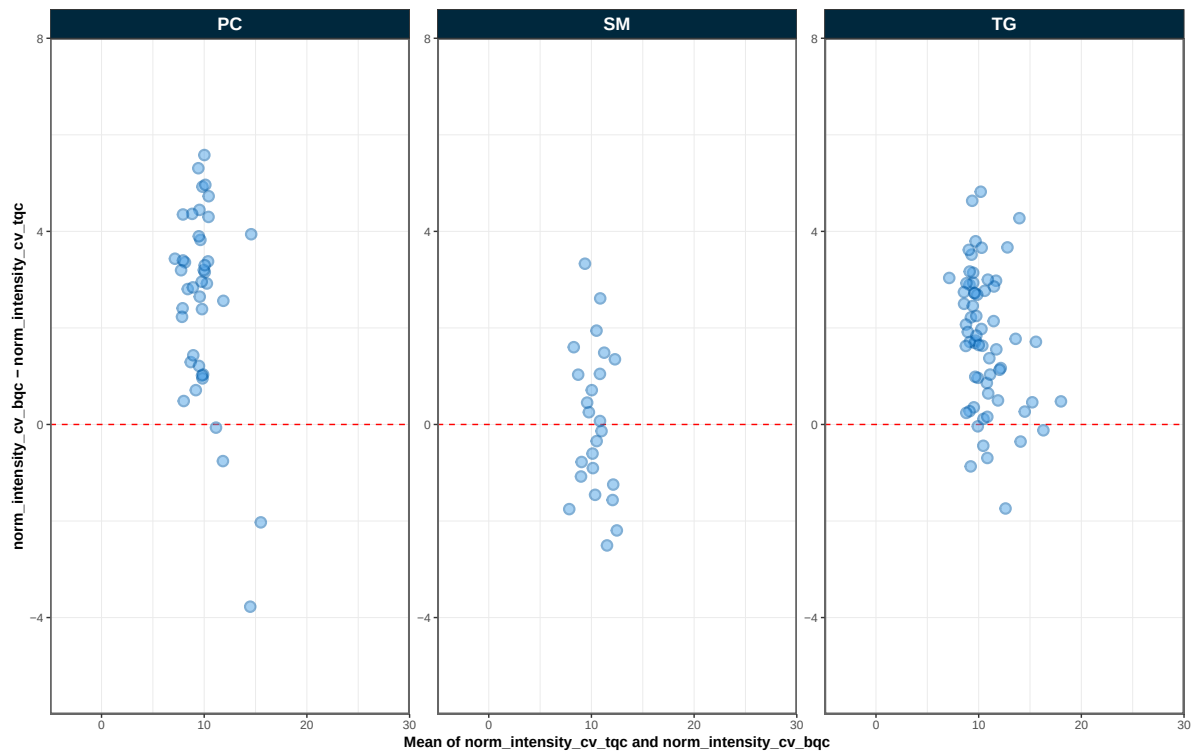
Compare the variability in BQCs vs TQCs may be useful to understand the source of variability in the dataset. If the variability in BQCs is much higher than in TQCs, this may suggest that the sample processing is a major contributor to overall variability. If the variability in TQCs and BQCs is similar, this may suggest that technical variability is a major contributor to overall variability. We first look at two sphingomyelins (SM), phosphatidylcholines (PC), and triglycerides (TG). PC and TG species consistently show higher variability in BQCs than in TQCs, suggesting that sample processing, i.e., lipid extraction was a major contributor to

overall variability, whereby it is unclear what factor(s) in the sample processing contributed to this variability.

SMs on the other hand show a similar variability in BQCs and TQCs, suggesting that the sample processing less contributed to the overall variability. This may be due to the fact that SMs are generally more abundant and thus less affected by variability in extraction efficiency.

```
mexp_temp <- calc_qc_metrics(mexp, use_robust_cv = TRUE, use_batch_medians = FALSE)
mexp_temp@metrics_qc <- mexp_temp@metrics_qc |>
  filter(str_detect(feature_class, "SM|TG$|^PC$"))

fig5e <- plot_qcmetrics_comparison(
  mexp_temp,
  plot_type = "diff",
  y_shared = FALSE,
  "norm_intensity_cv_tqc",
  "norm_intensity_cv_bqc",
  log_scale = FALSE,
  equality_line = TRUE,
  facet_by_class = TRUE,
  point_size = 1.3,
  font_base_size = 5,
  x_lim = c(0, 25),
  y_lim = c(-4, 6) )+
  theme(
    strip.text = ggplot2::element_text(size = 6),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    legend.position.inside = c(0.9, 0.10),
    axis.text = element_text(size = 4),
    #axis.title = element_blank(),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_text(size = 6*0.7),
    legend.key.size = unit(6 *0.7, "pt"))
fig5e
```



Next, we look at all lipid classes...

```
mexp_temp <- calc_qc_metrics(mexp, use_robust_cv = TRUE, use_batch_medians = FALSE)

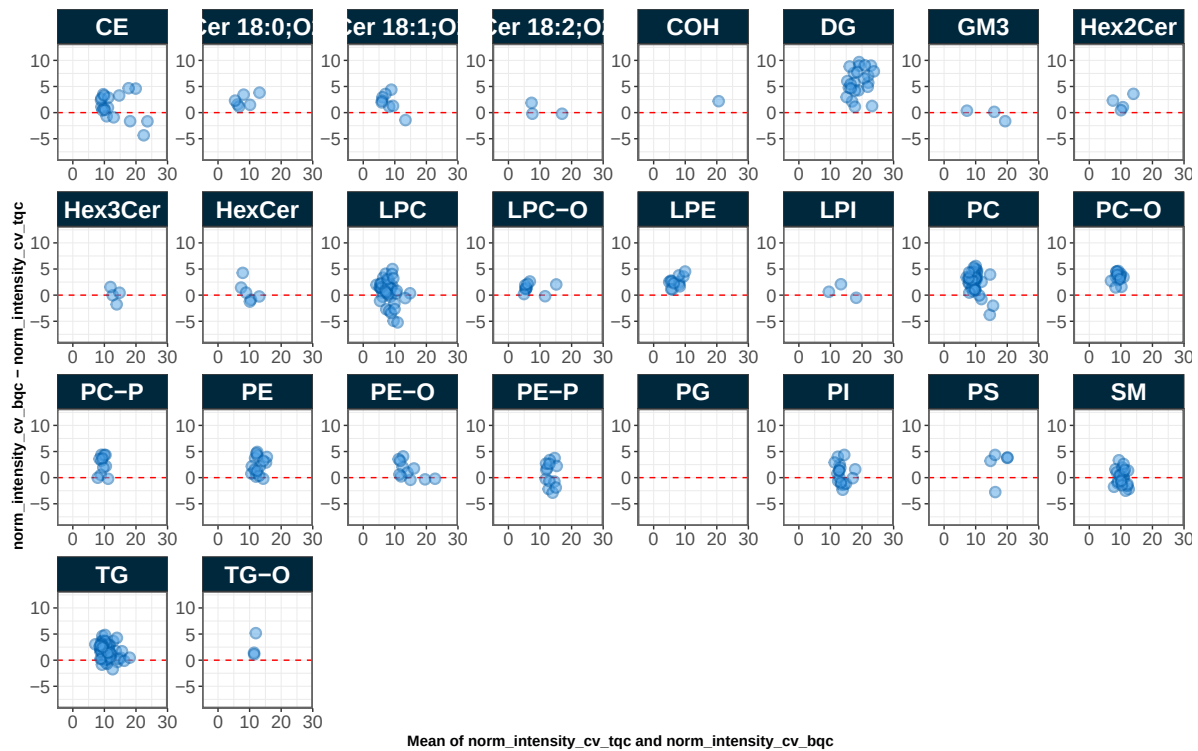
extfig7 <- plot_qcmetrics_comparison(
  mexp_temp,
  plot_type = "diff",
  y_shared = FALSE,
  "norm_intensity_cv_tqc",
  "norm_intensity_cv_bqc",
  log_scale = FALSE,
  equality_line = TRUE,
  facet_by_class = TRUE, cols_page = 8,
  point_size = 1.3,
  font_base_size = 5,
  x_lim = c(0, 25),
  y_lim = c(-6, 10)) +
  theme(
    strip.text = ggplot2::element_text(size = 8),
    #aspect.ratio = 0.9,
    legend.position = "inside",
```

```

legend.position.inside = c(0.9, 0.10),
axis.text = element_text(size = 6),
#axis.title = element_blank(),
legend.direction = "vertical",
legend.text = element_text(size = 9*0.7),
legend.title = element_text(size = 9*0.7),
legend.key.size = unit(6 *0.7, "pt")

```

extfig7



18. Response curves

```

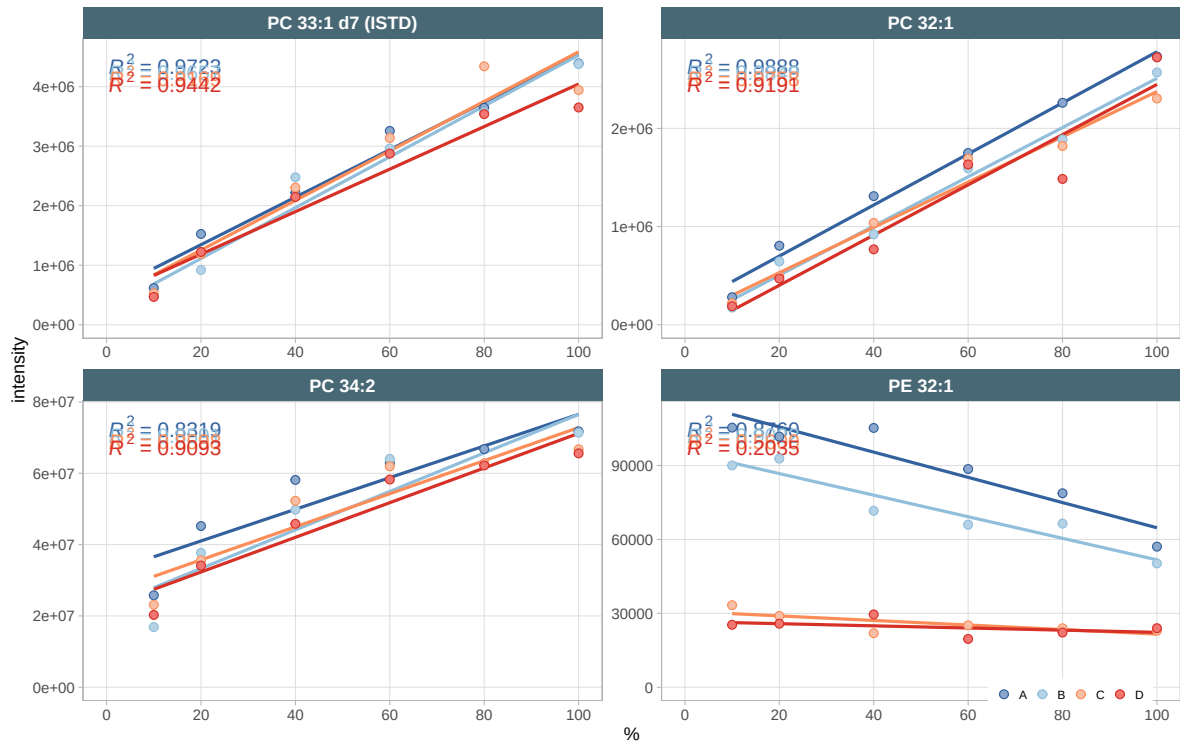
sel_species <- c("PC 33:1 d7 (ISTD)", "PC 32:1", "PC 34:2", "PE 32:1")
mexp_temp <- mexp
mexp_temp@dataset <- mexp_temp@dataset %>%
  filter(feature_id %in% sel_species) %>%
  mutate(feature_id = factor(feature_id, levels = sel_species)) %>%
  arrange(feature_id)

```

```

fig5f <- plot_responsecurves(
  data = mexp_temp,
  variable = "intensity",
  filter_data = FALSE, font_base_size = 6, line_width = 0.5, point_size = 1.2,
  include_feature_filter = sel_species,
  output_pdf = FALSE, path = "response-curves.pdf",
  show_progress = FALSE,
  cols_page = 2, rows_page = 2, return_plots = T)[[1]] +
  theme(
    panel.grid.minor = element_blank(),
    legend.position = "inside",
    legend.direction = "horizontal",
    legend.text = element_text(size = base_font_size*0.6),
    legend.title = element_blank(),
    legend.key.size = unit(base_font_size *0.6, "pt"),
    strip.text = element_text(size = 6),
    legend.position.inside = c(0.9, 0.01))
#> Registered S3 methods overwritten by 'ggpp':
#>   method          from
#> heightDetails.titleGrob ggplot2
#> widthDetails.titleGrob  ggplot2
#> Generating plots (1 page)... done!
fig5f

```



20. Feature filter

```
mexp_final <- filter_features_qc(
  data = mexp,
  recalc_metrics = TRUE,
  clear_existing = TRUE,
  use_batch_medians = TRUE,
  include_qualifier = FALSE,
  include_istd = FALSE,
  response.curves.selection = c(1,2),
  response.curves.summary = "mean",
  min.rsquare.response = 0.8,
  min.slope.response = 0.5,
  max.yintercept.response = 0.5,
  min.signalblank.median.spl.pblk = 10,
  min.intensity.median.spl = 100,
  max.cv.conc.bqc = 25,
  max.dratio.sd.conc.bqc = 0.75,
  max.prop.missing.conc.spl = 100,
```



```

    features.to.keep = c("CE 20:4", "CE 22:5", "CE 22:6", "CE 16:0", "CE 18:0")
  )
#> Calculating feature QC metrics - please wait...
#> ! The following features were forced to be retained despite not meeting filtering criteria
#> v
New feature QC filters were defined: 316 of 412 quantifier features meet QC criteria (not in

```

21. Feature filter Results

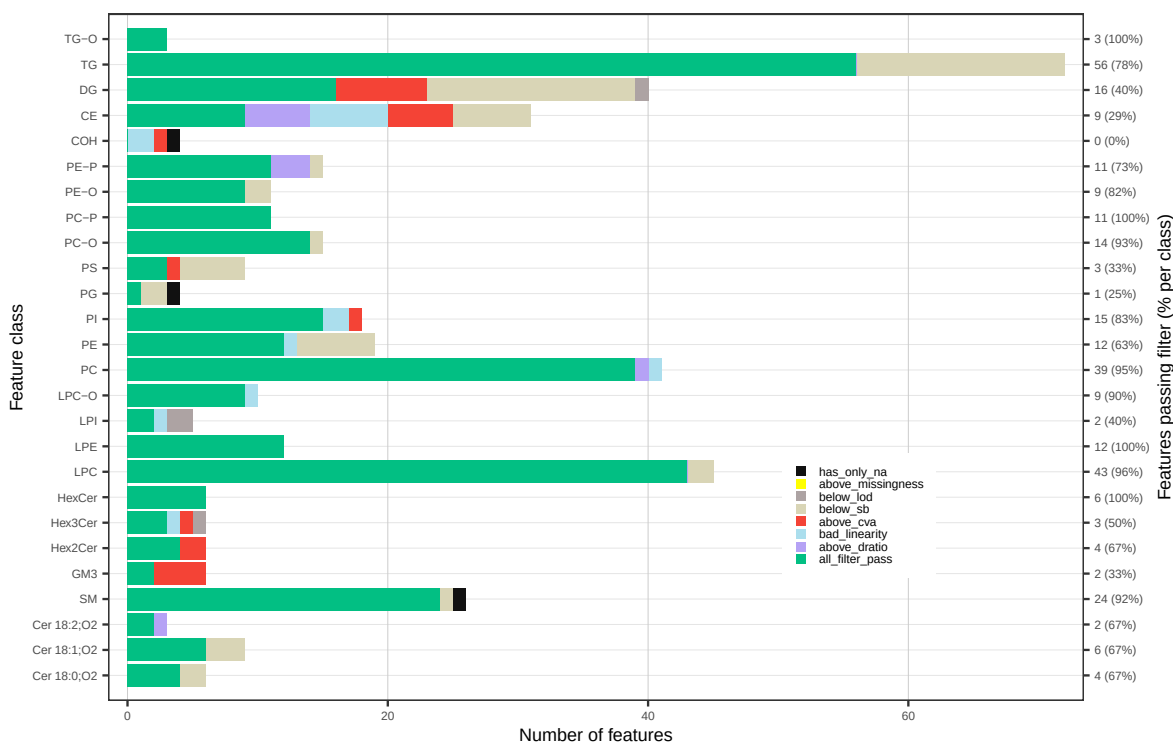
```

lipidCat_order <- c("Cer 18:0;02", "Cer 18:1;02", "Cer 18:2;02", "SM", "GM3", "Hex2Cer", "H
    "LPC", "LPE", "LPI", "LPC-0", "PC", "PE", "PI", "PG", "PS",
    "PC-0", "PC-P", "PE-0", "PE-P", "COH", "CE", "DG", "TG", "TG-0")

mexp_final@metrics_qc$feature_class <-
  factor(mexp_final@metrics_qc$feature_class, rev(lipidCat_order))
mexp_final@metrics_qc <- mexp_final@metrics_qc |> arrange(feature_class)

fig5h <- plot_qc_summary_byclass(mexp_final) +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    axis.text = element_text(size = 4),
    axis.title = element_text(size = 6, face = "plain"),
    axis.text.y.right = element_text(size = 4, face = "plain"),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_blank(),
    legend.key.size = unit(6 *0.7, "pt"),
    legend.position.inside = c(0.77, 0.27))
fig5h

```

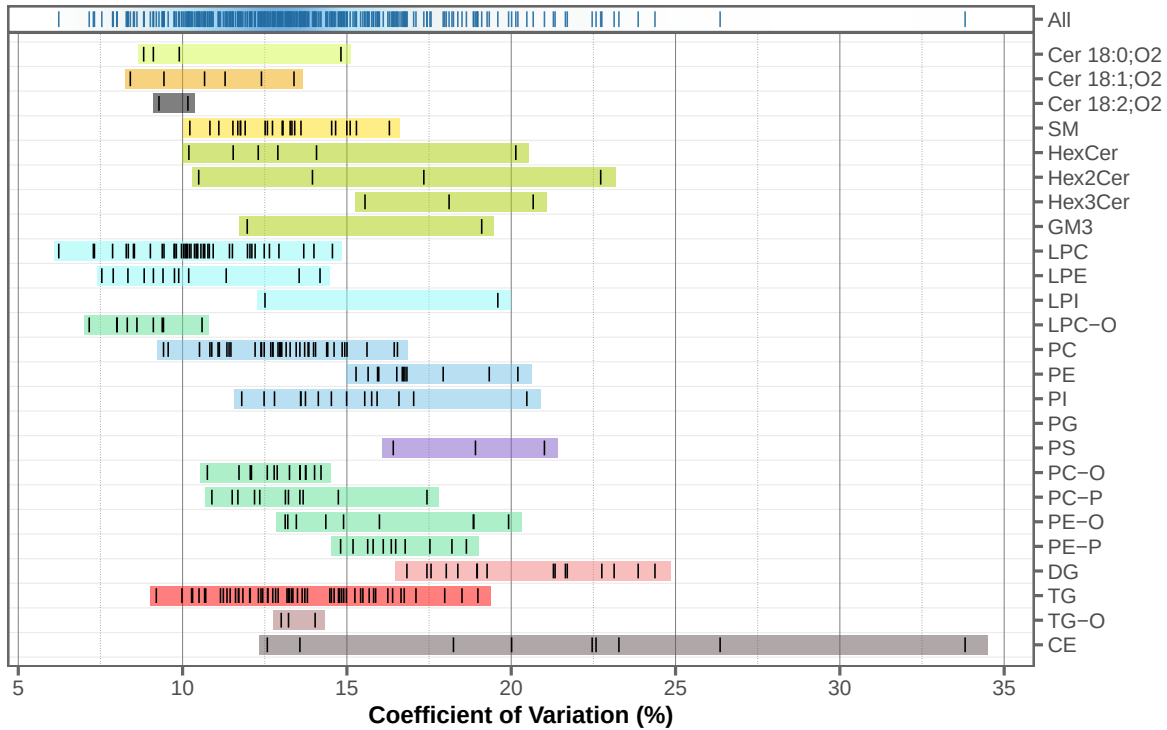


22. Overview analytical variability

The plot below provides an overview of the analytical variability of the filtered dataset, showing the distribution of coefficients of variation (CVs) in the BQCs for all features. We also see that the cholesteryl esters (CE), which we included in the final dataset despite failing QC, show a higher variability. In subsequent statistical analyses and interpretation, it will be important to consider that these CE species were forced to be included in the dataset.

```
plot_abundanceprofile(
  data = mexp_final,
  use_qc_metrics = TRUE,
  log_scale = FALSE,
  filter_data = TRUE,
  variable = "conc_cv_bqc",
  density_strip = TRUE,
  qc_types = "SPL",
  #analysis_range = c(1,4000),
  show_sum = FALSE,
  #x_lim = c(6.5, 7.5),
```

```
x_label = NA,
feature_map = "lipidomics")
```



23. Feature filter Venn

```
fig5ex <- plot_qc_summary_overall(mexp_final)

# fig5h <- fig5hi[[1]] +
#   theme(
#     strip.text = ggplot2::element_text(size = 5),
#     #aspect.ratio = 0.9,
#     legend.position = "none",
#     axis.text = element_text(size = 4),
#     axis.title = element_blank(),
#     # legend.direction = "vertical",
#     # legend.text = element_text(size = 6*0.7),
#     # legend.title = element_text(size = 6*0.7),
#     # legend.key.size = unit(6 *0.7, "pt")
```

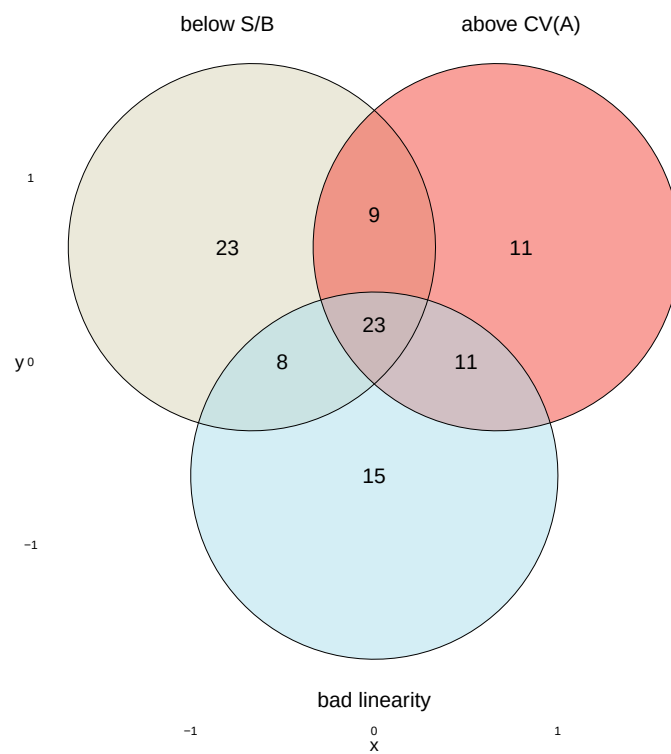
```

#      )
#
# fig5h

fig5g <- fig5ex[[2]] +
  theme(
    strip.text = ggplot2::element_text(size = 5),
    #aspect.ratio = 0.9,
    legend.position = "inside",
    axis.text = element_text(size = 4),
    axis.title = element_text(size = 6),
    legend.direction = "vertical",
    legend.text = element_text(size = 6*0.7),
    legend.title = element_text(size = 6*0.7),
    legend.key.size = unit(6 *0.7, "pt"))

```

fig5g

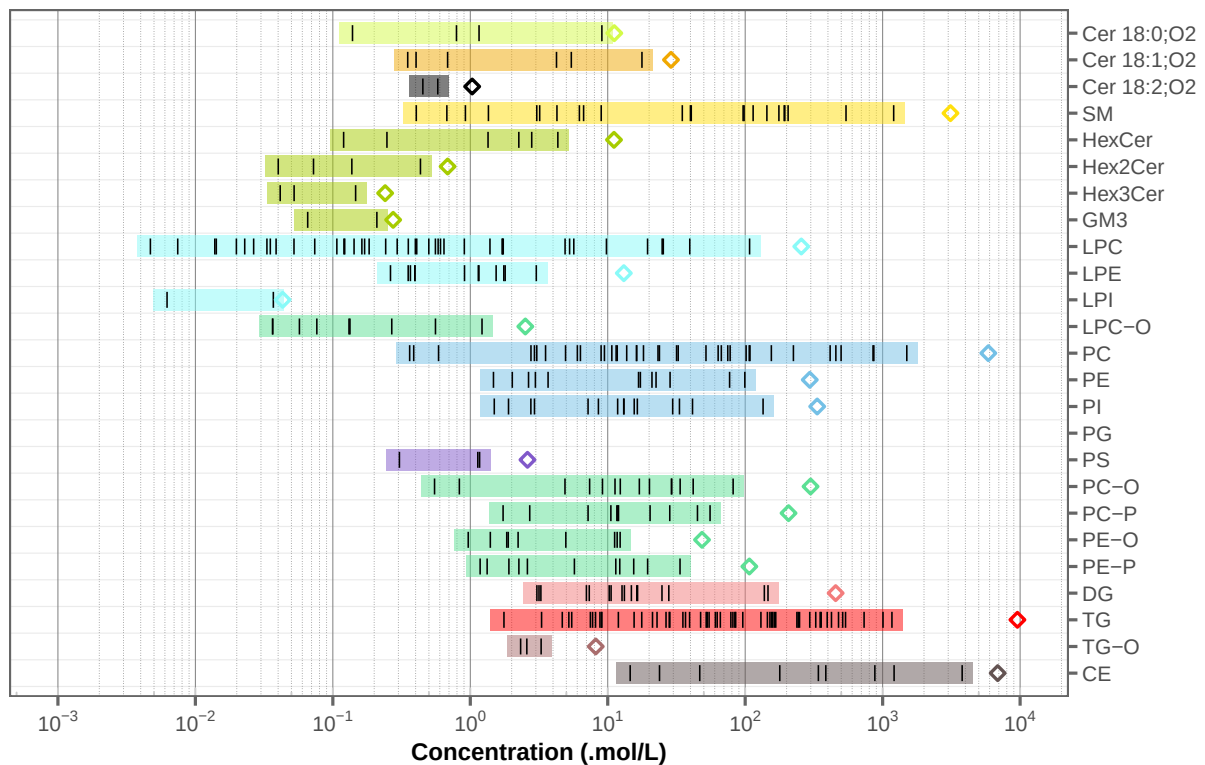


24. Lipidome Profile

As a final overview, the feature concentration profile of the filtered dataset is shown below. Validating the concentrations, for example, those of the most abundant species, the summed concentrations per lipid class, or the ratios between lipid classes, against in-house reference values or literature data helps ensure that the quantification is within expected ranges and that no major quantification errors are present.

```
fig5i <- plot_abundanceprofile(  
  data = mexp_final,  
  log_scale = TRUE,  
  variable = "conc",  
  filter_data = TRUE,  
  qc_types = "SPL",  
  #x_lim = c(-6, 2),  
  x_label = NA,  
  feature_map = "lipidomics")
```

fig5i



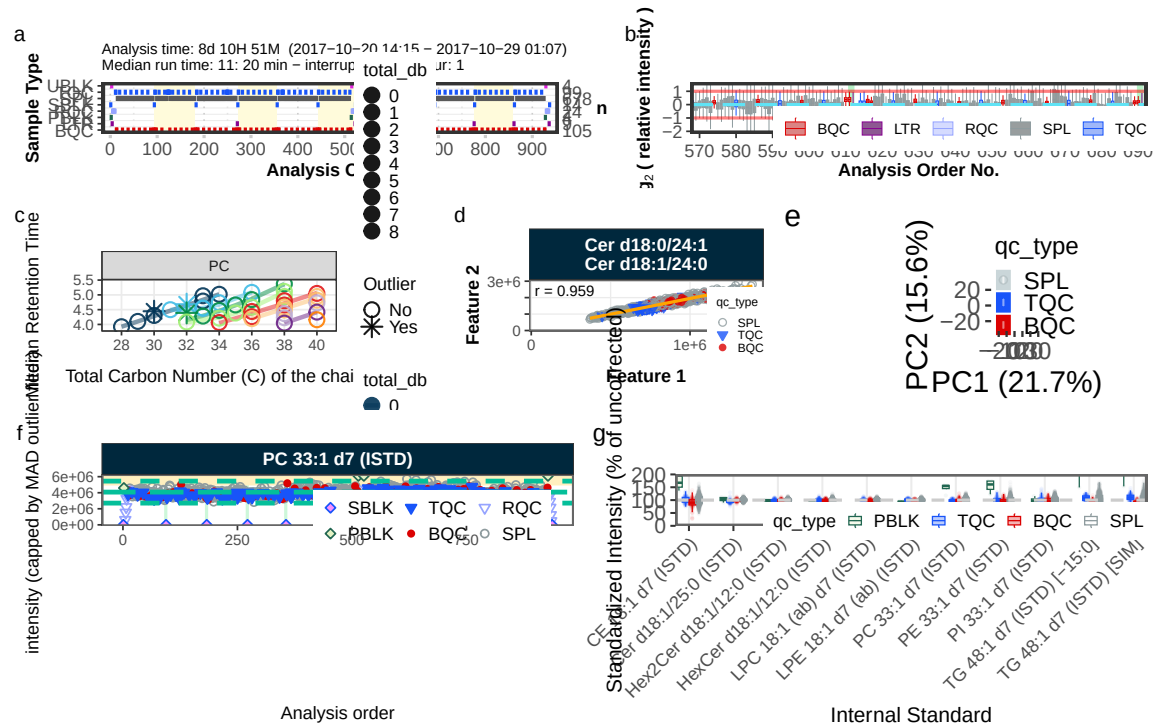
25. Exporting final dataset

```
save_dataset_csv(  
  data = mexp,  
  path = "./output/sperfect_UNFILTERED_RAW-feature_conc_uM_20250518a.csv",  
  variable = "conc_raw",  
  qc_types = "SPL",  
  include_qualifier = FALSE,  
  filter_data = FALSE)  
#> v Conc_raw values for 678 analyses and 412 features have been exported to './output/sperf
```

Assembling figures for the manuscript

1. Assembling Figure 4

```
fig4 <- ((fig4a | fig4b) / ((fig4c | fig4d | fig4e) +  
  plot_layout( widths = c(0.9,0.9,1), heights = c(0.8,1,1))) / (fig4f | fig4g)) +  
  plot_layout(heights = c(1,1,1)) +  
  plot_annotation(tag_levels = 'a') +  
  theme(plot.tag = element_text(size = 12, face = "bold"),  
    strip.placement = 'inside')  
fig4
```



```
ggsave("output/fig4.png", fig4, width = 180, height = 180, units = "mm", dpi = 300)
ggsave("output/fig4.pdf", fig4, width = 180, height = 180, units = "mm", dpi = 300)
```

2. Assembling Figure 5

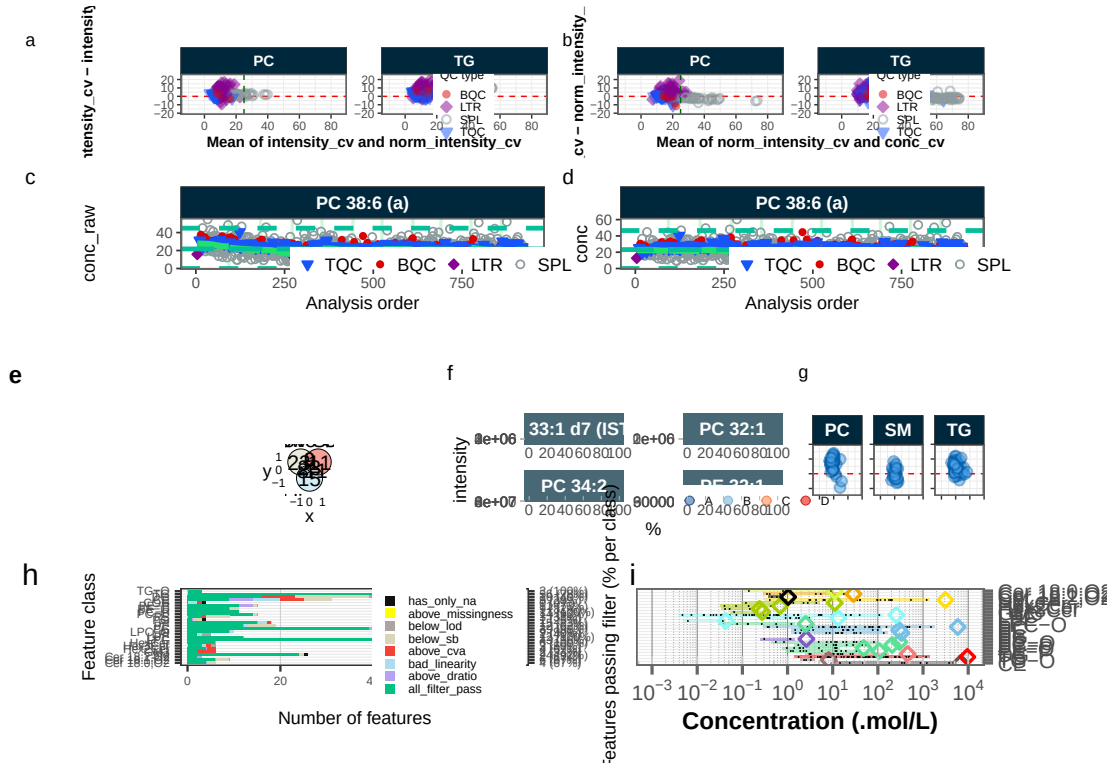
```
layout <- "
AABB
CCDD
EEFF
"

fig5 <- (fig5a | fig5b) /
  #(p_fig2c | p_fig2d) /
  (fig5c | fig5d) /
  ((fig5g | fig5f | (fig5e + theme(axis.text = element_blank(), axis.title = element_blank())) /
    plot_layout(widths = c(1.2, 1.2, 0.80))) /
  (fig5h | fig5i) +
  plot_layout(heights = c(0.8, 1, 1, 1.5)) +
```

```

plot_annotation(tag_levels = 'a') +
  theme(plot.tag = element_text(size = 10, face = "bold"),
        strip.placement = 'inside')
fig5

```



```

ggsave("output/fig5.png", fig5, width = 180, height = 240, units = "mm", dpi = 300)
ggsave("output/fig5.pdf", fig5, width = 180, height = 240, units = "mm", dpi = 300)
#> Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
#> conversion failure on 'Concentration (mol/L)' in 'mbsToSbcs': for (U+03BC)

```

3. Assembling Extended Figure 2

```

# include fig4b again as extfig2a for better comparison of RLAs before and after normalization
extfig2a <- fig4b

extfig2 <- extfig2a / extfig2b / extfig2c / extfig2d + plot_layout(ncol = 1, heights = c(1,
  plot_annotation(tag_levels = 'a') +
  theme(plot.tag = element_text(size = 19, face = "bold"),

```

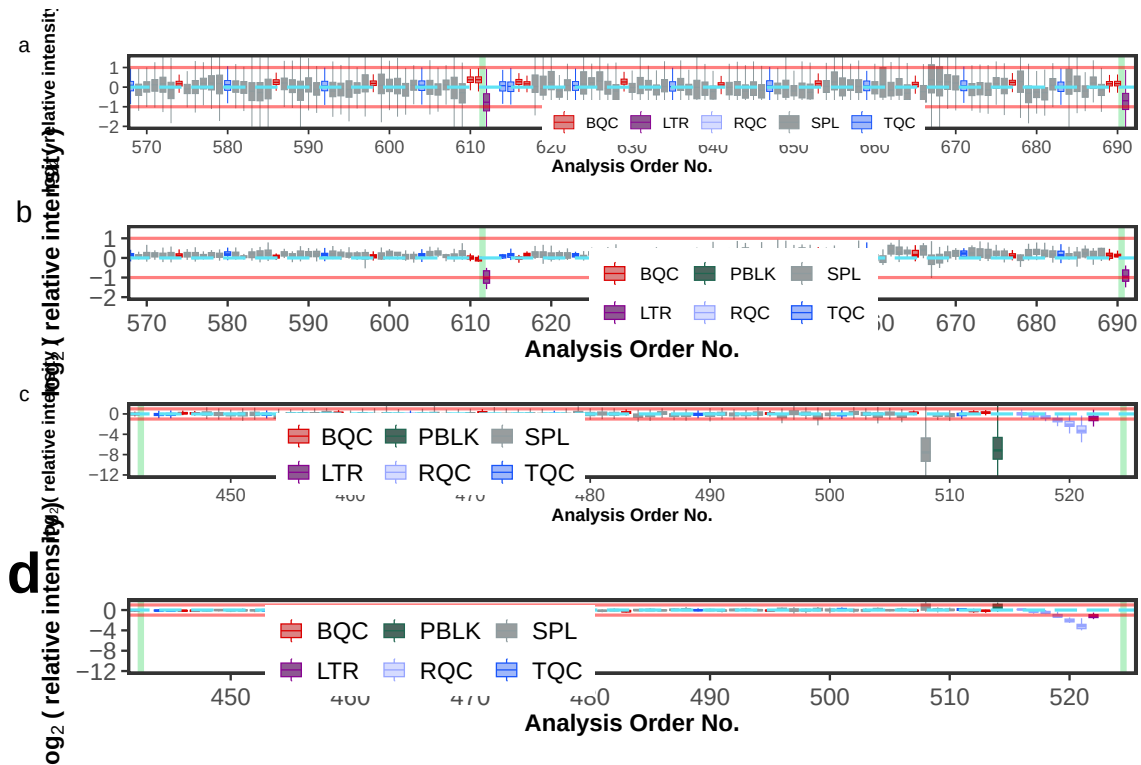


```

strip.placement = 'inside')

ggsave(extfig2, filename = "output/ExtendedFig-2_RLAs.pdf",
       width = 180, height = 250, units = "mm", dpi = 300)
extfig2

```



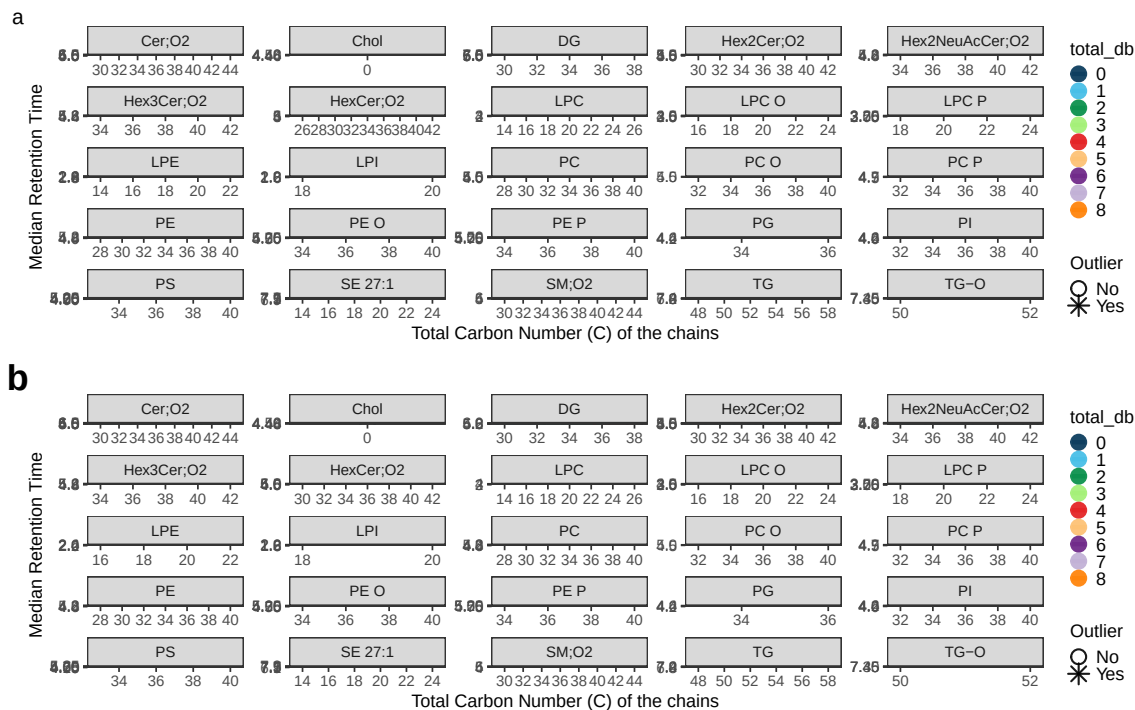
3. Assembling Extended Figure 3

```

extfig3 <- extfig3a / extfig3b +
  plot_annotation(tag_levels = 'a')+
  theme(plot.tag = element_text(size = 12, face = "bold"),
        strip.placement = 'inside')

ggsave(extfig3, filename = "output/ExtendedFig-3.png",
       width = 180, height = 250, units = "mm", dpi = 300)
extfig3

```

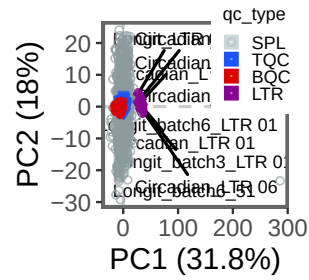


4. Assembling Extended Figure 4

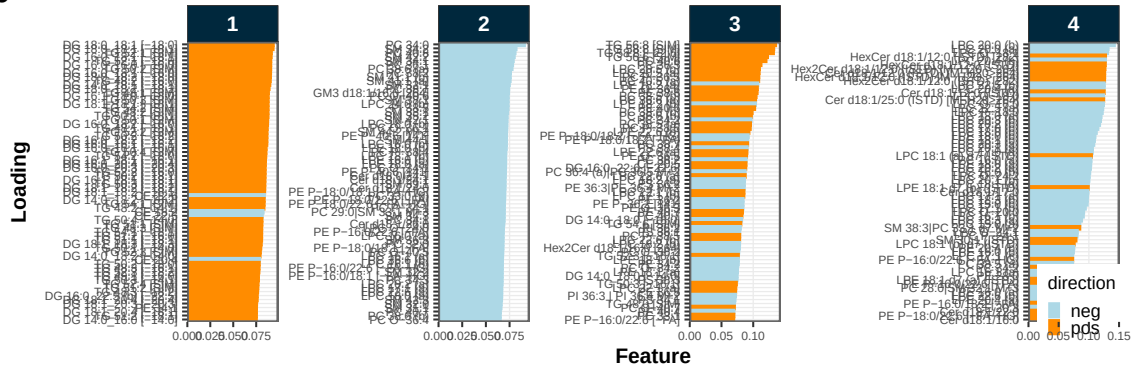
```
extfig4 <- extfig4a / extfig4b + plot_layout(ncol = 1, heights = c(1,1.5)) +
  plot_annotation(tag_levels = 'a') +
  theme(plot.tag = element_text(size = 12, face = "bold"),
        strip.placement = 'inside')

ggsave(
  extfig4, filename = "output/ExtendedFig-4_PCAloding.png",
  width = 180, height = 250, units = "mm", dpi = 300)
ggsave(
  extfig4, filename = "output/ExtendedFig-4_PCAloding.pdf",
  width = 180, height = 250, units = "mm", dpi = 300)
extfig4
```

a



b



5. Saving Extended Figure 5

```
ggsave(extfig5[[1]], filename = "output/ExtendedData_Fig5_ISTDtrends.png",
       width = 180, height = 180, units = "mm", dpi = 300)
```

5. Assembling Extended Figure 6

```
extfig6 <- extfig6a / extfig6b / extfig6c + plot_layout(ncol = 1, heights = c(1,1,1)) +
  plot_annotation(tag_levels = 'a') +
  theme(plot.tag = element_text(size = 12, face = "bold"),
        strip.placement = 'inside')
ggsave(extfig6, filename = "output/ExtendedData_Fig6_CVnormbeforeafter.pdf",
       width = 160, height = 270, units = "mm", dpi = 600)
extfig6
```



6. Saving Extended Figure 7

```
ggsave(extfig7, filename = "output/ExtendedData_Fig7_CV_TQC_BQC_V2.pdf",
       width = 160, height = 200, units = "mm", dpi = 600)
```

Supplementary Figures of Manuscript

Runscatter plots

```
plot_runscatter(mexp,
  variable = "intensity",
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  show_reference_lines = TRUE,
  ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
```

```

show_trend = FALSE,
point_size = 1.2,
base_font_size = 6,
cols_page = 3,
rows_page = 3,
cap_outliers = TRUE,
reference_sd_shade = FALSE,
#batch_zebra_stripe = TRUE,
show_progress = FALSE,
output_pdf = TRUE,
path = "output/suppl-fig1_runscatter_area.pdf", return_plot = FALSE)
#> Saving plots to pdf (53 pages) done!
plot_runscatter(mexp,
  variable = "conc_raw",
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  show_reference_lines = TRUE,
  ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
  show_trend = TRUE,
  point_size = 1.2,
  base_font_size = 6,
  cols_page = 3,
  rows_page = 3,
  cap_outliers = TRUE,
  reference_sd_shade = FALSE,
  #batch_zebra_stripe = TRUE,
  show_progress = FALSE,
  output_pdf = TRUE,
  path = "output/suppl-fig2_runscatter_conc-raw.pdf", return_plot = FALSE)
#> Saving plots to pdf (53 pages) done!
plot_runscatter(mexp,
  variable = "conc",
  qc_types = c("SPL", "BQC", "TQC", "LTR"),
  show_reference_lines = TRUE,
  ref_qc_types = "SPL",
  reference_fill_color = "#111111",
  reference_k_sd = 3,
  show_trend = TRUE,
  point_size = 1.2,
  base_font_size = 6,
  cols_page = 3,

```

```
rows_page = 3,  
cap_outliers = TRUE,  
reference_sd_shade = FALSE,  
#batch_zebra_stripe = TRUE,  
show_progress = FALSE,  
output_pdf = TRUE,  
path = "output/suppl-fig3_runscatter_conc-final.pdf", return_plot = FALSE)  
#> Saving plots to pdf (53 pages) done!
```